

HCDE 511

Information Visualization

Winter 2026

9

Week

Week 1

Intro
Data
encoding &
mapping

2

Graphical
excellence
& integrity
Critiquing

3

Fundamen-
tal graphs
Working
with data

4

Tableau &
dashboard
design

Visual
perception

5

Visual
perception,
cont.
Tasks &
inter-
actions

6

Collabora-
tive
critique

7

Multi-
variate
analysis

8

Trees &
Networks

Project
check-in

9

Different-
iation
Story-
telling,
Animation,
Context

Project
feedback

10

Final pre-
sentations

Finals

A1 Good/bad

A2 Static vis

A3 Exploratory analysis

Project proposal

Rev. prop.

Final project deliverable

Final project writeup

Next Week

Final project presentations!
Final projects!

12-15 min presentation

Goals & data

User questions / needs

Your process

Interactive dashboard / visualizations

Critical evaluation

3-5 mins Q&A



Storytelling

Break

Animation + Context

Break

~~Project feedback: group to group~~

Project Group Work Time

Storytelling

**Guiding
users**

**Providing
context**

**Creating
understanding**

Storytelling

**Guiding
users**

Providing
context

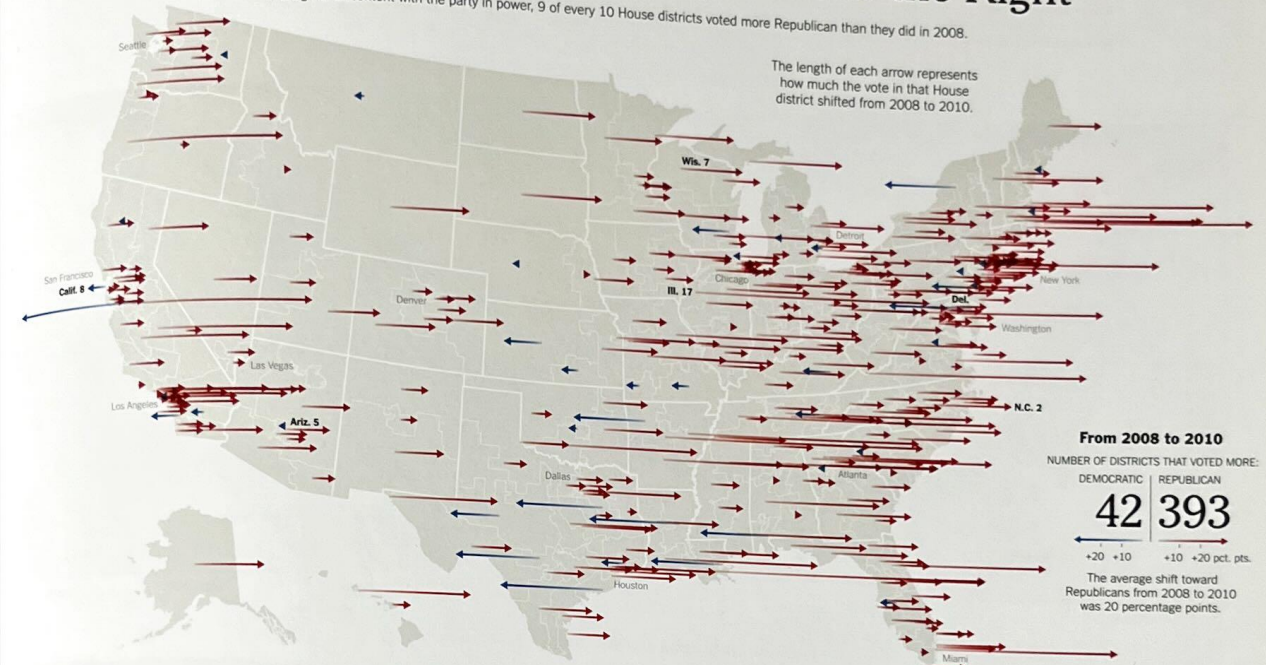
Creating
understanding

Guiding users
through static
visualizations

Selection of
data + views

Districts Across the Country Shift to the Right

In a sign of discontent with the party in power, 9 of every 10 House districts voted more Republican than they did in 2008.



While Republicans increased their share of the vote in **California 8**, Nancy Pelosi's lead still increased in the absence of a strong third-party candidate.

Arizona 5 shifted right about 20 points — enough to switch the seat to Republicans. David Schweikert defeated the Democrat he lost to in 2008.

The shift in **Wisconsin 7** was about average for an open race. Here, Sean P. Duffy, a Republican district attorney, won by 8 percentage points.

One of the largest shifts was in **Illinois 17**, where Republican Bobby Schilling, a pizza business owner, beat Phil Hare, a two-term Democratic incumbent.

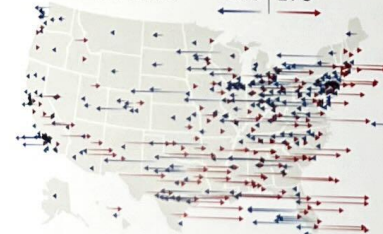
Only a few districts voted more Democratic. In **Delaware**, the shift helped John Carney defeat Glen Urquhart for the seat held by Michael N. Castle since 1993.

Renee Ellmers delivered one of the Republican Party's narrowest gains in **North Carolina 2**, a district that Democrats won by 36 percentage points in 2008.

Previous Shifts

From 2002 to 2004

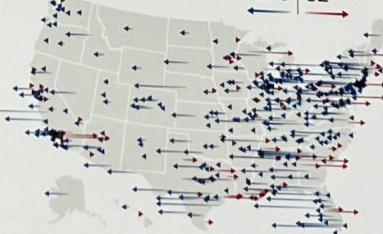
NO. OF DISTRICTS: 260 | 175



With President George W. Bush up for his second term in a close race against John Kerry, most districts voted more Democratic.

From 2004 to 2006

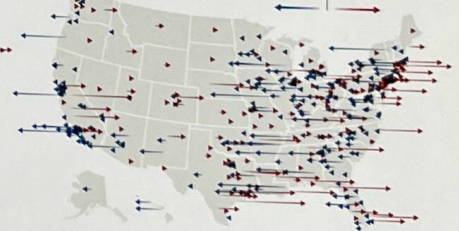
353 | 82



Discontent with President Bush, the war in Iraq and the handling of Hurricane Katrina helped the Democrats take control of the House and Senate.

From 2006 to 2008

245 | 190



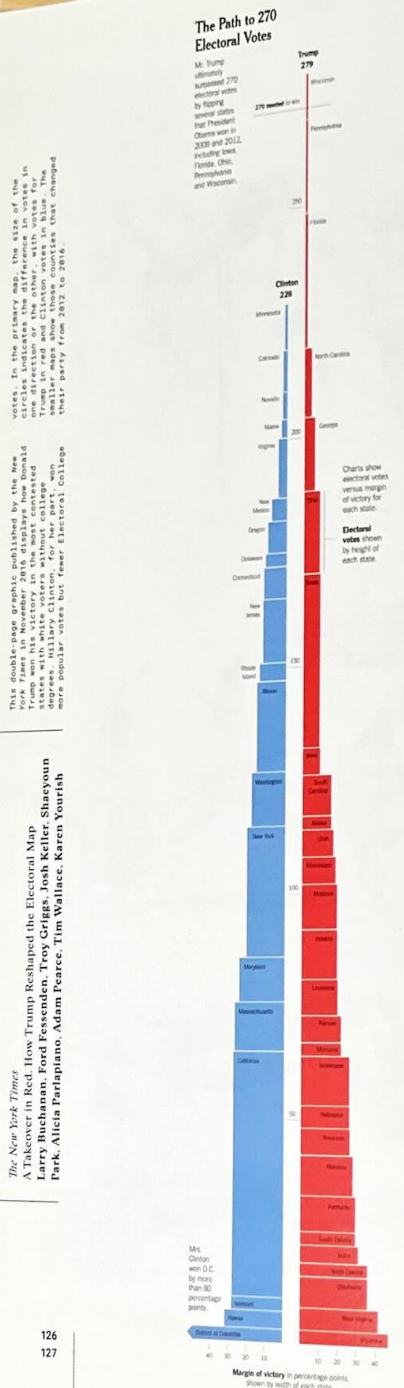
A majority of districts, riding the wave of Barack Obama's campaign, moved toward the Democrats, but the shift was less widespread than in 2006.

Guiding users through static visualizations

Layout/ composition

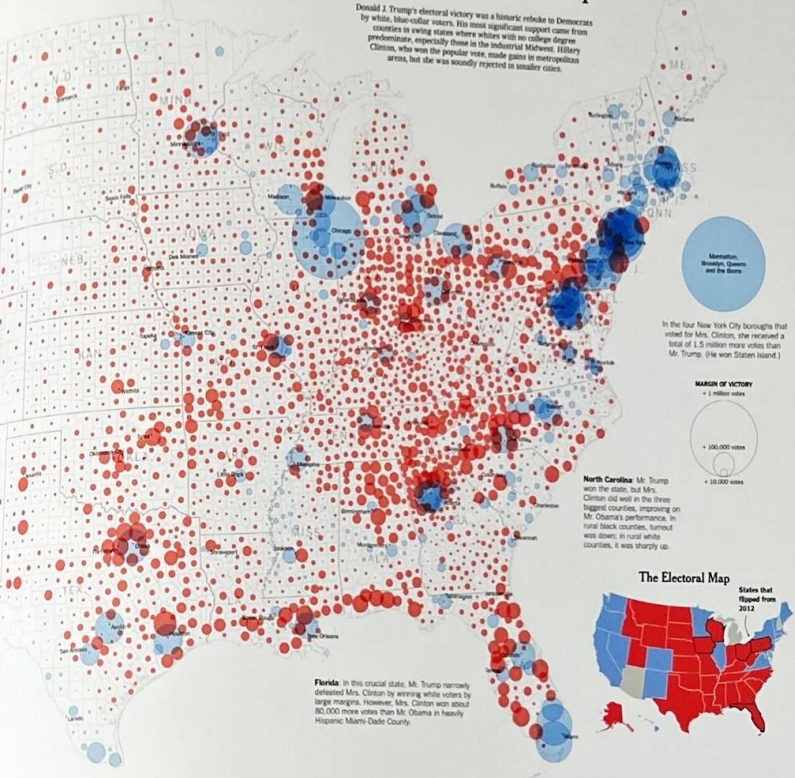
The New York Times
A Takeover in Red: How Trump Reshaped the Electoral Map
Larry Buchanan, Ford Fessenden, Troy Griggs, Josh Keller, Sharyoun Park, Alicia Parlapiano, Adam Pearce, Tim Wallace, Karen Yourish

This double-page graphic published by the New York Times in November 2016 displays how Donald Trump's victory in the 2016 presidential election was achieved. The graphic shows the electoral map, the popular vote, and the electoral college vote. It also includes a map of the United States showing the distribution of the electoral college votes by state. The graphic is a static visualization of the 2016 election results.



Popular vote
59,760,117 59,547,218
47.7% 47.5%
Electoral vote
228 279

How Trump Reshaped The Electoral Map



The Vote by County

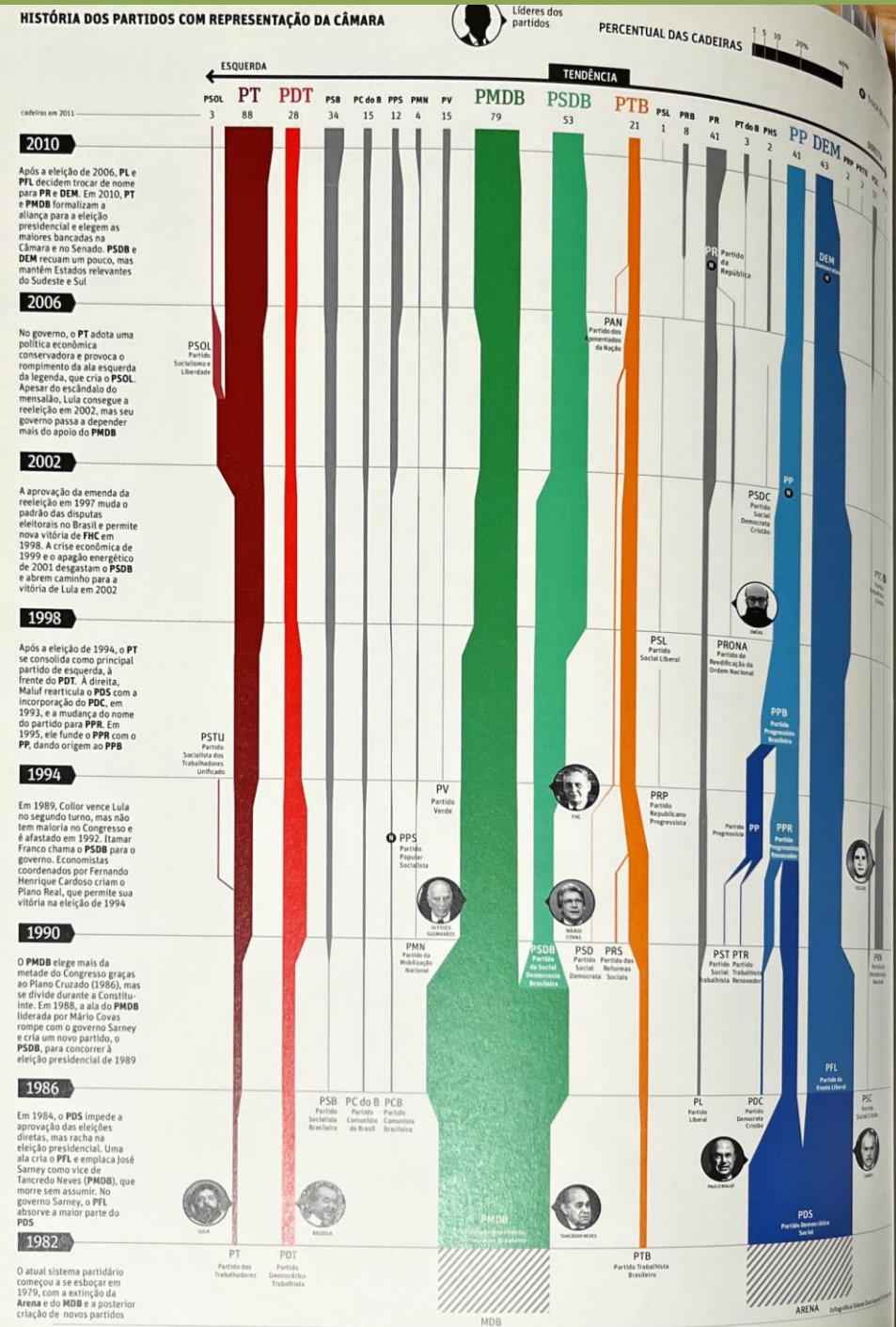


How the States Voted

States won by Donald J. Trump				States won by Hillary Clinton			
State	Clinton	Trump	Other	State	Clinton	Trump	Other
Alabama	100%	100%	0%	California	100%	100%	0%
Alaska	100%	100%	0%	Colorado	100%	100%	0%
Arizona	100%	100%	0%	Connecticut	100%	100%	0%
Arkansas	100%	100%	0%	District of Columbia	100%	100%	0%
California	100%	100%	0%	Florida	100%	100%	0%
Colorado	100%	100%	0%	Georgia	100%	100%	0%
Connecticut	100%	100%	0%	Hawaii	100%	100%	0%
District of Columbia	100%	100%	0%	Idaho	100%	100%	0%
Florida	100%	100%	0%	Illinois	100%	100%	0%
Georgia	100%	100%	0%	Indiana	100%	100%	0%
Hawaii	100%	100%	0%	Iowa	100%	100%	0%
Idaho	100%	100%	0%	Kansas	100%	100%	0%
Illinois	100%	100%	0%	Kentucky	100%	100%	0%
Indiana	100%	100%	0%	Louisiana	100%	100%	0%
Iowa	100%	100%	0%	Maine	100%	100%	0%
Kansas	100%	100%	0%	Maryland	100%	100%	0%
Kentucky	100%	100%	0%	Massachusetts	100%	100%	0%
Louisiana	100%	100%	0%	Michigan	100%	100%	0%
Maine	100%	100%	0%	Minnesota	100%	100%	0%
Maryland	100%	100%	0%	Mississippi	100%	100%	0%
Massachusetts	100%	100%	0%	Missouri	100%	100%	0%
Michigan	100%	100%	0%	Montana	100%	100%	0%
Minnesota	100%	100%	0%	Nebraska	100%	100%	0%
Mississippi	100%	100%	0%	Nevada	100%	100%	0%
Missouri	100%	100%	0%	New Hampshire	100%	100%	0%
Montana	100%	100%	0%	New Jersey	100%	100%	0%
Nebraska	100%	100%	0%	New Mexico	100%	100%	0%
Nevada	100%	100%	0%	New York	100%	100%	0%
New Hampshire	100%	100%	0%	North Carolina	100%	100%	0%
New Jersey	100%	100%	0%	North Dakota	100%	100%	0%
New Mexico	100%	100%	0%	Ohio	100%	100%	0%
New York	100%	100%	0%	Oklahoma	100%	100%	0%
North Carolina	100%	100%	0%	Oregon	100%	100%	0%
North Dakota	100%	100%	0%	Pennsylvania	100%	100%	0%
Ohio	100%	100%	0%	Rhode Island	100%	100%	0%
Oklahoma	100%	100%	0%	South Carolina	100%	100%	0%
Oregon	100%	100%	0%	South Dakota	100%	100%	0%
Pennsylvania	100%	100%	0%	Tennessee	100%	100%	0%
Rhode Island	100%	100%	0%	Texas	100%	100%	0%
South Carolina	100%	100%	0%	Utah	100%	100%	0%
South Dakota	100%	100%	0%	Vermont	100%	100%	0%
Tennessee	100%	100%	0%	Virginia	100%	100%	0%
Texas	100%	100%	0%	Washington	100%	100%	0%
Utah	100%	100%	0%	West Virginia	100%	100%	0%
Vermont	100%	100%	0%	Wisconsin	100%	100%	0%
Virginia	100%	100%	0%	Wyoming	100%	100%	0%
Washington	100%	100%	0%				
West Virginia	100%	100%	0%				
Wisconsin	100%	100%	0%				
Wyoming	100%	100%	0%				

Guiding users through static visualizations

Sequence + Flow

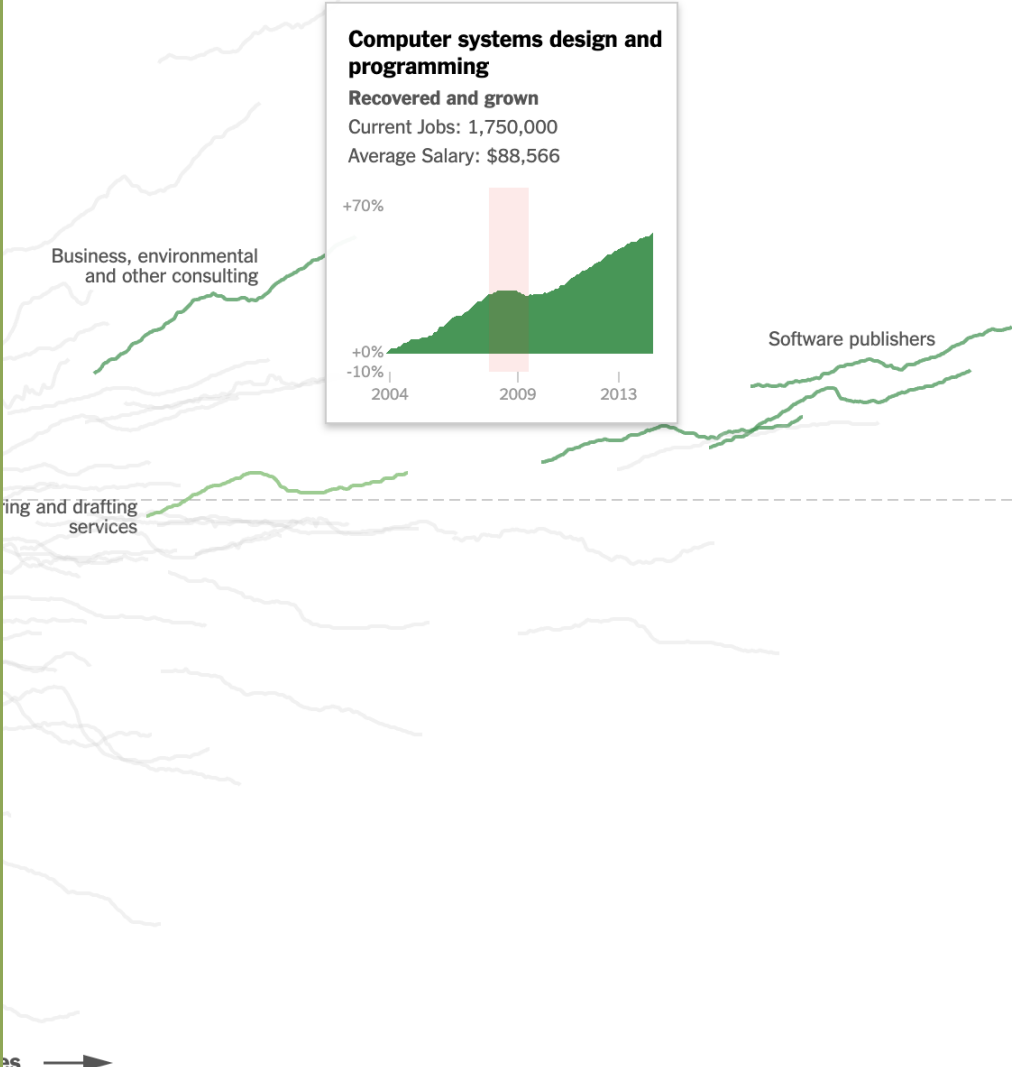


Guiding users through static visualizations

Highlighting

More Bad — and Good — Jobs

Americans often lament the quality of jobs today, and some low-paying industries — like fast food , where annual average pay is less than \$22,000 — are growing. But so are some high-paying sectors, such as consulting , computing and biotech . [NEXT »](#)



Guiding users
through
dynamic
visualizations

Sequential
reveal

Animations

Scrollytelling

The web we want

The dark side of Guardian comments

As part of a series on the rising global phenomenon of online harassment, the Guardian commissioned research into the 70m comments left on its site since 2006 and discovered that of the 10 most abused writers eight are women, and the two men are black. Hear from three of those writers, explore the data and [help us host better conversations online](#)

by [Becky Gardiner](#), [Mahana Mansfield](#), [Ian Anderson](#), [Josh Holder](#), [Daan Louter](#) and [Monica Ulmanu](#)

Comments allow readers to respond to an article instantly, asking questions, pointing out errors, giving new leads. At their best, comment threads are thoughtful, enlightening, funny: online communities where readers interact with journalists and others in ways that enrich the Guardian's journalism.

But at their worst, they are something else entirely.

[The Guardian](#) was not the only news site to turn comments on, nor has it been the only one to find that some of what is written "below the line" is crude, bigoted or just vile. On all news sites where comments appear, too often things are said to journalists and other readers that would be unimaginable face to face - the Guardian is no exception.

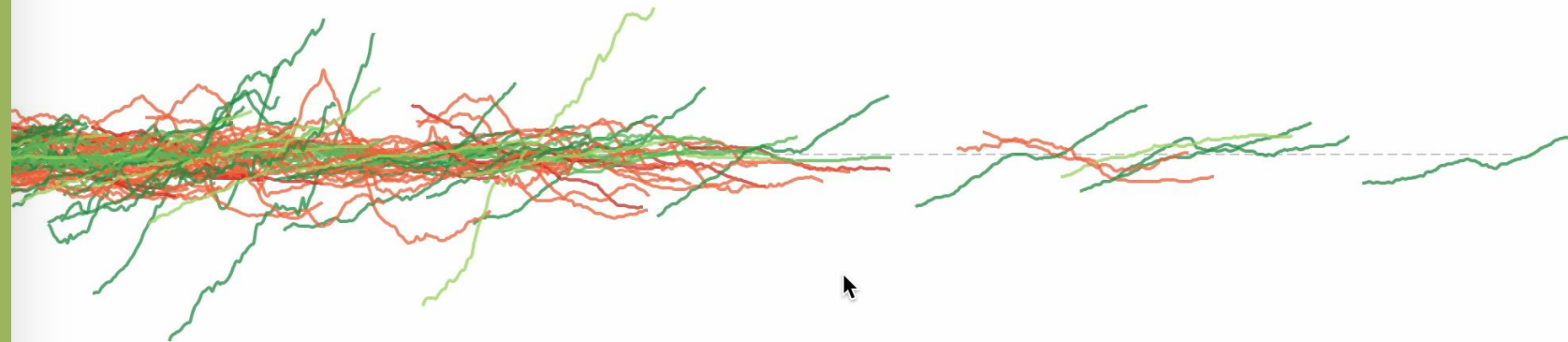
Guiding users through dynamic visualizations

Interactions
Pre-selection
of filters,
queries, views

How the Recession Reshaped the Economy, in 255 Charts

By JEREMY ASHKENAS and ALICIA PARLAPIANO Updated: JUNE 6, 2014

End of the Great Recession, the economy has finally regained the nine million jobs lost in 2003. But the recovery has not been equal. Each line below shows how the number of jobs in a particular industry over the past 10 years. Scroll down to see how the recession reshaped the nation's job market, industry by industry.



SCROLL



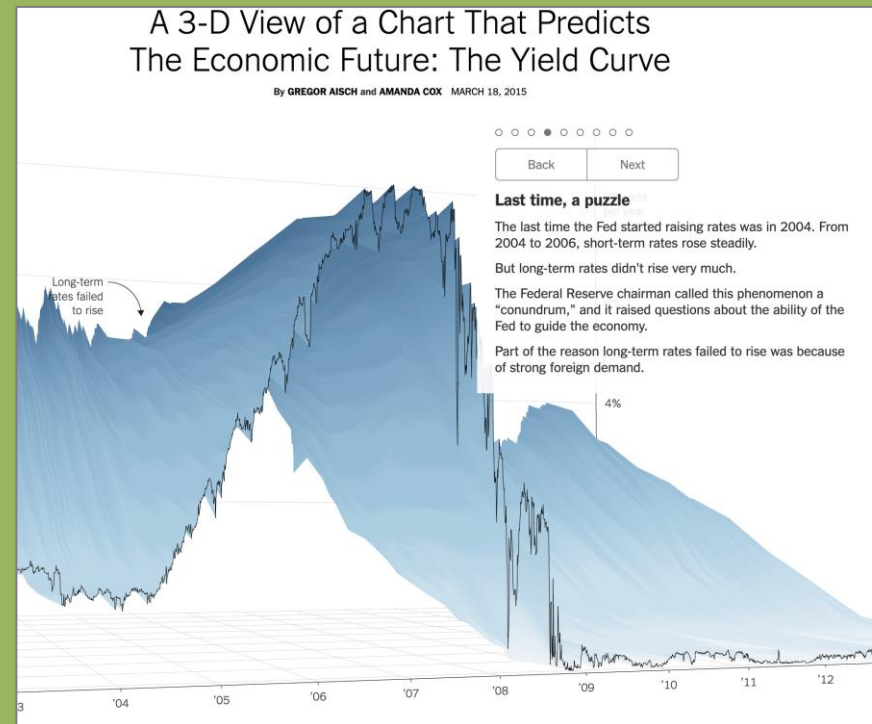
Storytelling

Guiding
users

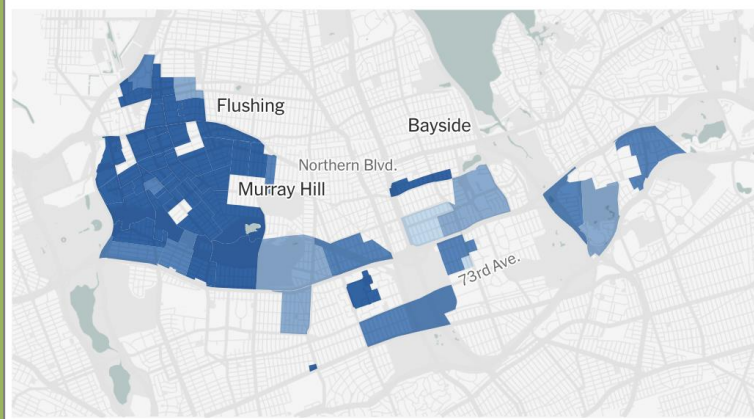
Providing
context

Creating
understanding

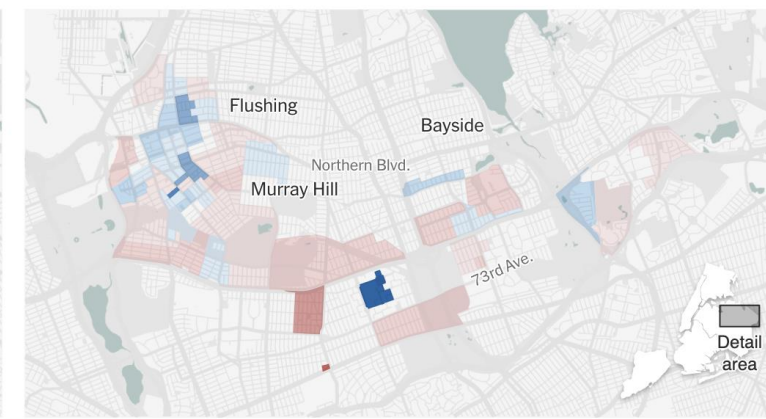
Captioning



Flushing, 2018



Flushing, 2022



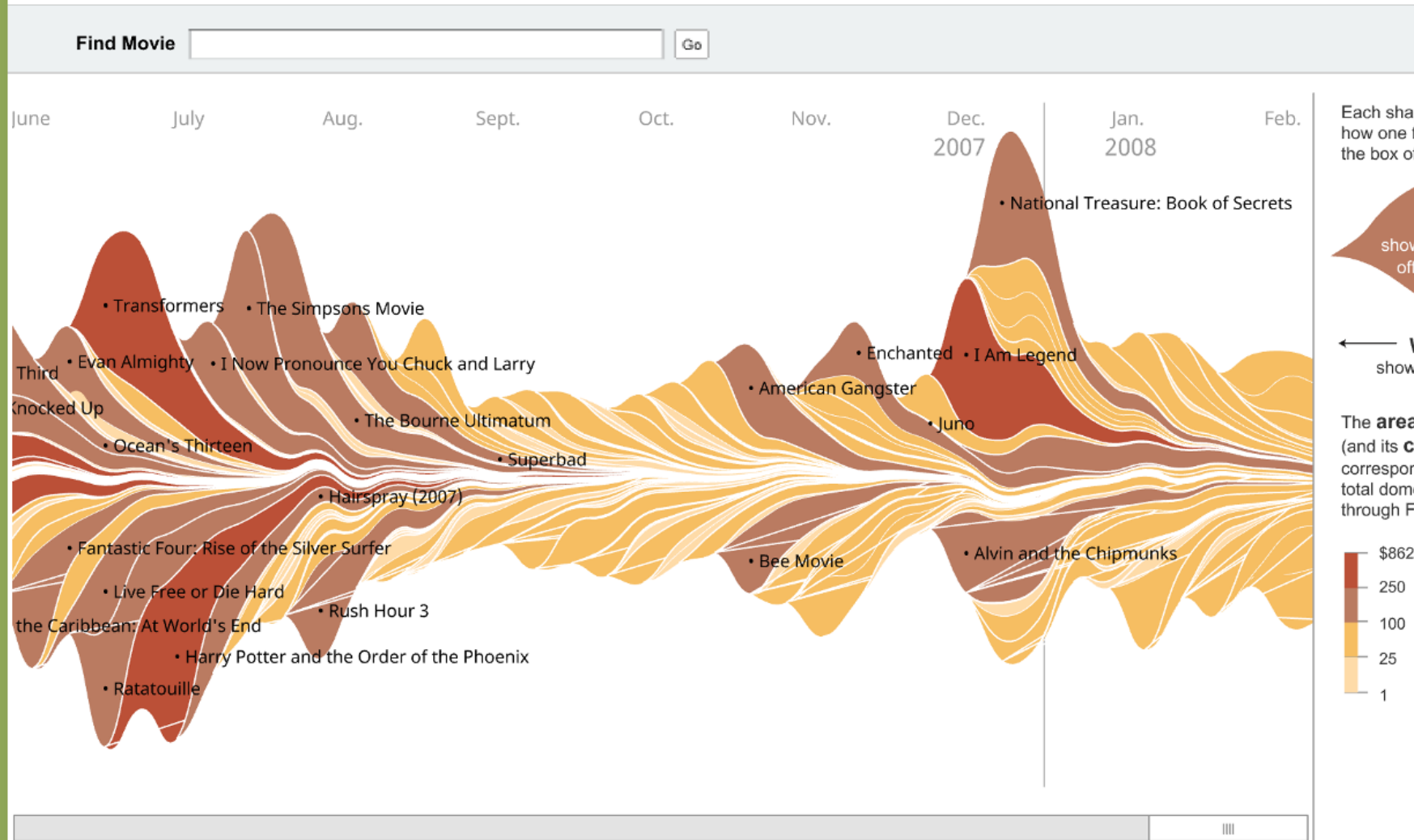
Providing context

Introductory text

February 23, 2008

The Ebb and Flow of Movies: Box Office Receipts 1986 — 2008

Summer blockbusters and holiday hits make up the bulk of box office revenue each year, while contenders for the Oscars tend to attract smaller audiences that build over time. Here's a look at how movies have fared at the box office, after adjusting for inflation.



Sources: Baseline StudioSystems; Box Office Mojo

Mathew Bloch, Lee Byron, Shan Carter and

Providing
context

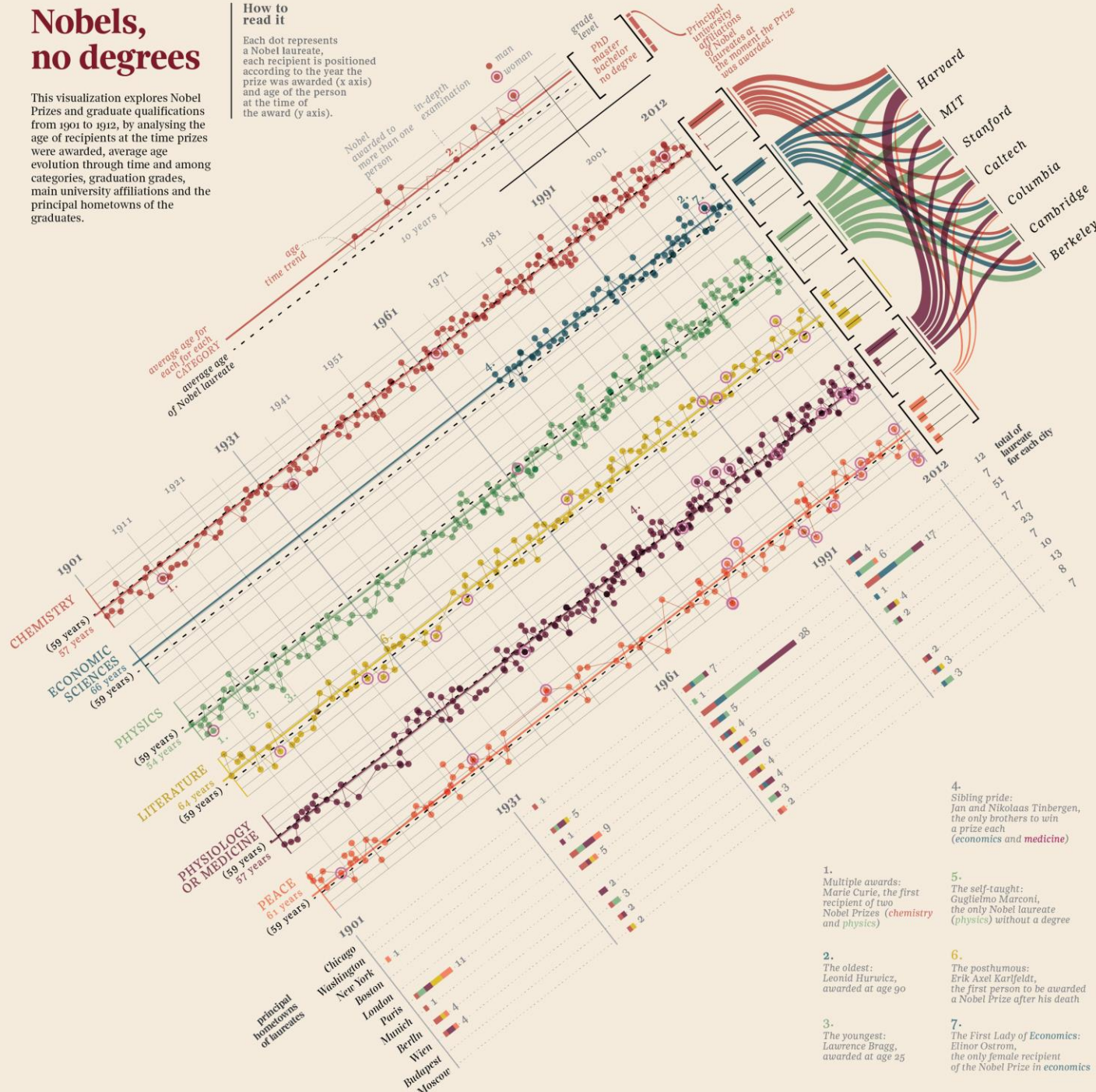
Annotations
Related events

Nobels, no degrees

This visualization explores Nobel Prizes and graduate qualifications from 1901 to 1912, by analysing the age of recipients at the time prizes were awarded, average age evolution through time and among categories, graduation grades, main university affiliations and the principal hometowns of the graduates.

How to read it

Each dot represents a Nobel laureate, each recipient is positioned according to the year the prize was awarded (x axis) and age of the person at the time of the award (y axis).



Illustrations/ pictures + data

Columna vertebral

Fluor peroteo

Retina

Tapetum lucidum

Pupila

Luz

Excelente visión nocturna
Una membrana reflectante (Tapetum lucidum) en el fondo del ojo concentra la luz en el campo focal de la retina, permitiendo desparar y ver con pocas condiciones de luz.

Paracen totalmente negro, pero de cerca se pueden apreciar las manchas.

En la selva, sus músculos direccionados al frente hacen que los nervios del hocico detecten el movimiento y envíen la señal al cerebro.

Hueso mandibular

Musculos

Hueso Maxilar

Caninos

Mordedura letal
Ejerce una presión de 2.000 libras por pulgada cuadrada. Posee la mandíbula más grande del mundo animal en relación al tamaño de su cara.

Es capaz de rugir, gracias a su alargada y especialmente ancha laringe.

Es una especie casi amenazada. En la Cacha ecuatoriana la situación es más crítica que en el Oriente respecto a números de ejemplares debido a la fragmentación de su hábitat.

Es un carnívoro capaz de comer hasta 25 kilogramos de carne de una sola vez, ya que puede haber periodos en los que escasea el alimento.

Los robustos caninos posibilitan la entrada en la piel y huesos de sus presas con un menor requerimiento mandibular que si fueran más delgados.

Un tenaz cazador que prefiere emboscar a perseguir
Andan en solitario sobre amplios territorios que en el caso de las hembras llegan a medir 40 km² y en los machos el doble o más. Los espacios pueden superponerse entre sí, pero los animales suelen evitarse entre ellos. Utilizan rugidos, arrastres en los árboles, orina y heces para marcar y defender su zona. Su naturaleza evasiva y lo inaccesible de gran parte de su hábitat hacen que sea difícil de observar.

MÉTODO DE CAZA

1. Se desplaza sigilosamente escuchando, y aanchando la presa.

2. La emboscada puede incluir saltar dentro del agua para perseguir a la presa.

Permanece hasta un 60% de su tiempo en actividad.

Habita desde la selva tropical al campo abierto, muy rara vez es visto en zonas montañosas.

Campos de visión del Jaguar

No tan veloz, pero muy persuasivo

Cuerpo 115 km/h. El animal más veloz del mundo.

Jaguar 60 km/h. Puede avanzar 6 metros con carga cargada.

Autómvil 100 km/h. Límite de velocidad en Ecuador.

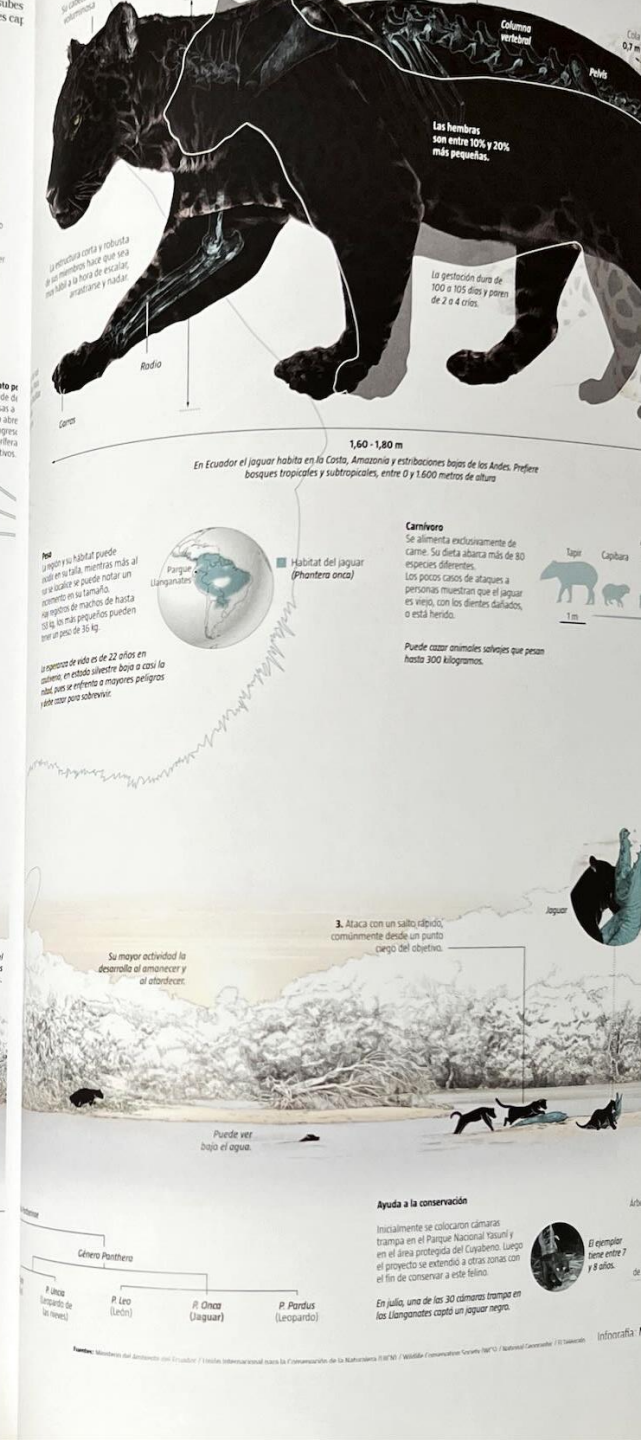
Humano 37 km/h. Usain Bolt, el hombre más veloz del mundo, corriendo los 100 metros planos.

El árbol familiar

Género Neofelis

N. Nebulosa (P. Nebulosa)

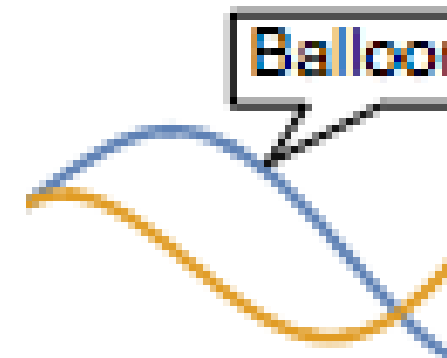
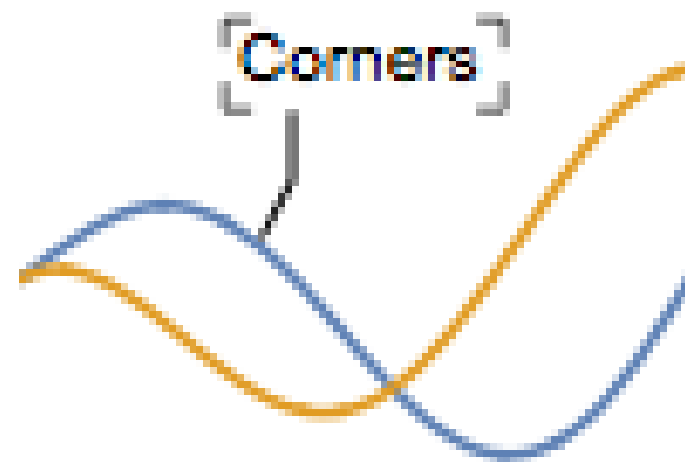
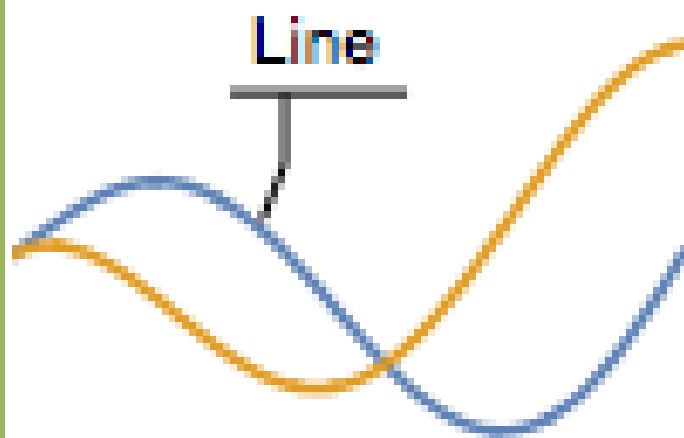
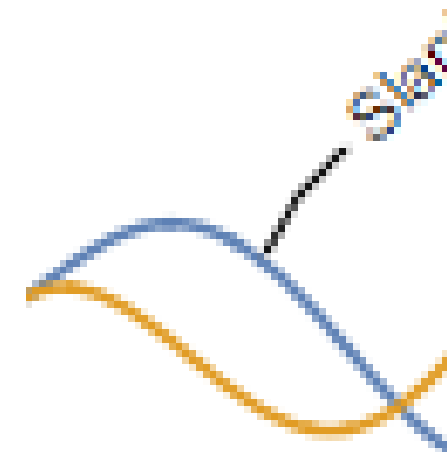
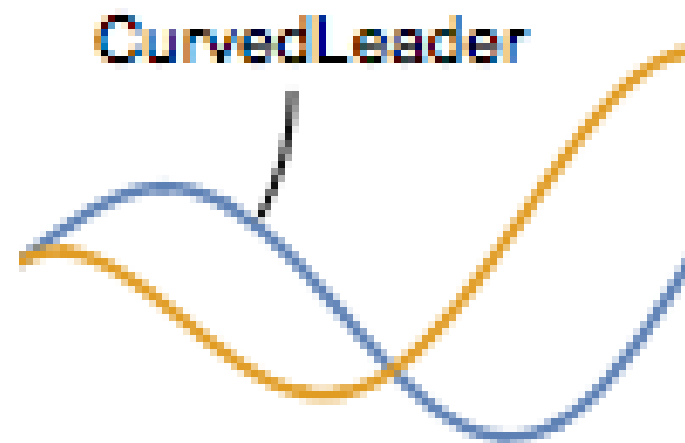
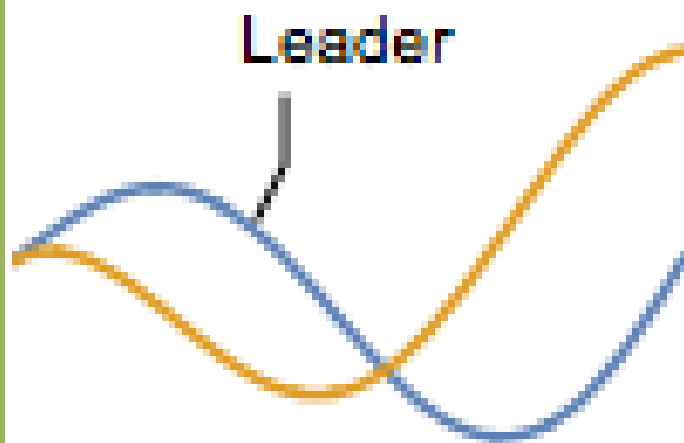
N. Dardi (P. Nebulosa de Bonelli)



Providing context

Callouts

Annotations



Storytelling

Guiding
users

Providing
context

Creating
understanding

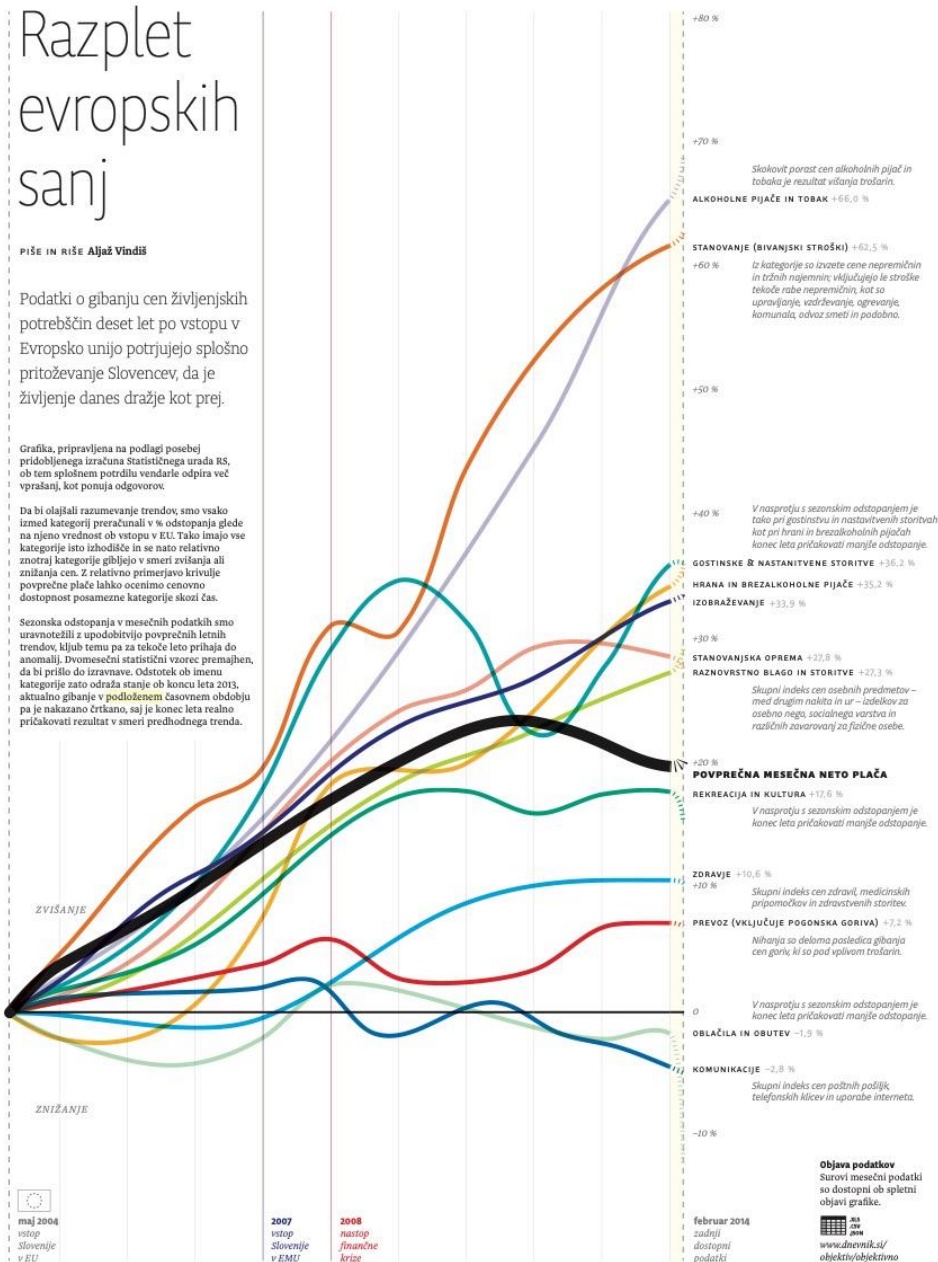
Legends (if done right)

PIŠE IN RIŠE Aljaž Vindiš

Grafika, pripravljena na podlagi posebej pridobljenega izražena Statističnega urada RS, ob tem splošnem potrdilo vendarle odpira več vprašanj, kot ponuja odgovorov.

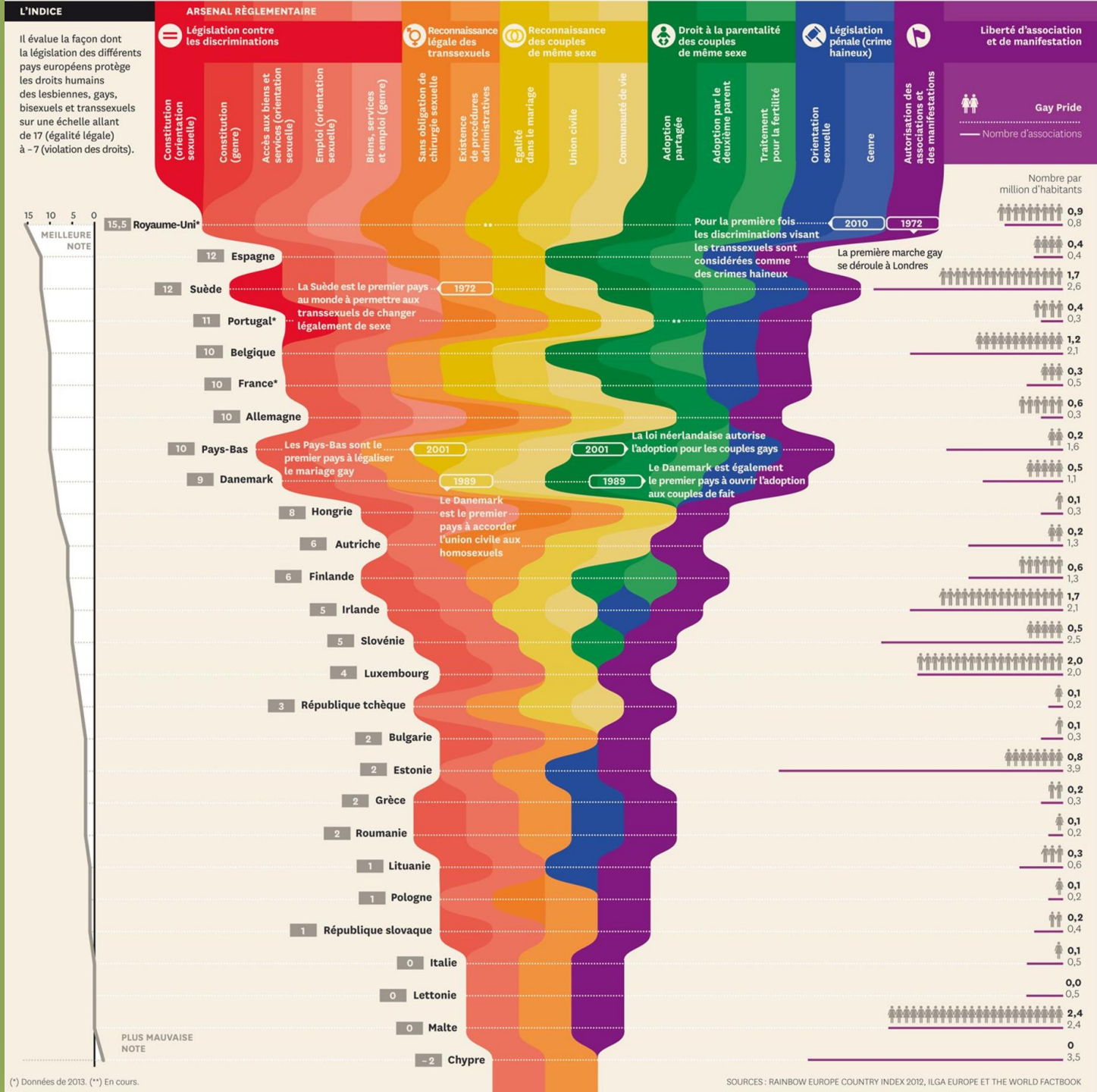
Da bi olajšali razumevanje trendov, smo vsako izmed kategorij preračunali v % odstopanja glede na njeno vrednost ob vstopu v EU. Tako imajo vse kategorije isto izhodišče in se nato relativno znotraj kategorije gibljejo v smeri zvišanja ali znižanja cen. Z relativno primerjavo krivulje povprečne plače lahko ocenimo cenovno dostopnost posamezne kategorije skozi čas.

Sezonska odstopanja v mesečnih podatkih smo uravnavili z upodobitvijo povprečnih letnih trendov, kljub temu pa za tekoče leto prihaja do anomalij. Dvomesecni statistični vzorec premajhen, da bi prišlo do izravnave. Odstopke ob imenu kategorije zato odraža stanje ob koncu leta 2013, aktualno gibanje v podloženem časovnem obdobju pa je nakazano črtno, saj je konec leta realno pričakovati rezultat v smeri predhodnega trenda.



Create understanding

Explanatory text



Visual vocabulary

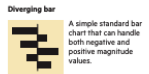
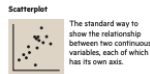
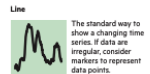
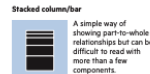
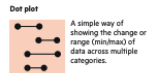
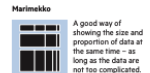
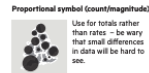
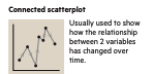
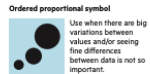
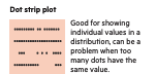
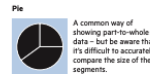
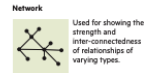
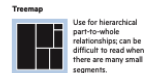
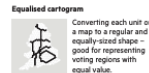
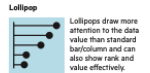
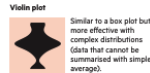
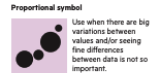
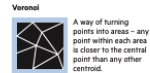
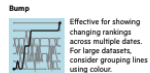
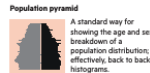
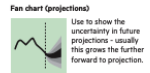
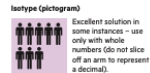
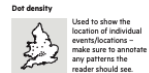
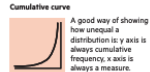
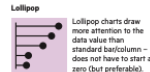
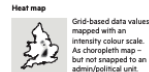
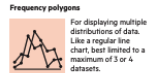
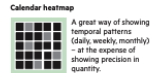
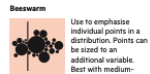
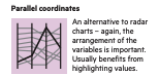
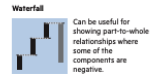
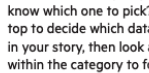
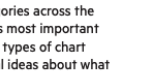
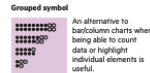
Designing with data

There are so many ways to visualise data - how do we know which one to pick? Use the categories across the top to decide which data relationship is most important in your story, then look at the different types of chart within the category to form some initial ideas about what might work best. This list is not meant to be exhaustive, nor a wizard, but is a useful starting point for making informative and meaningful data visualisations.

FT graphics: Alan Smith, Chris Campbell, Ian Borth, Liz Francis, Graham Parrish, Billy Ehrenberg-Shannon, Paul McCann, Martin Stobie
Inspired by the Graphic: Continuum by Jan Schwachlitz and Severino Bicecci



ft.com/vocabulary

Deviation	Correlation	Ranking	Distribution	Change over Time	Magnitude	Part-to-whole	Spatial	Flow
Emphasise variations (+/-) from a fixed reference point. Typically the reference point is zero but it can also be a target or a long-term average. Can also be used to show sentiment (positive/neutral/negative). Example FT uses Trade surplus/deficit, climate change	Show the relationship between two or more variables. Be mindful that, unless you tell them otherwise, many readers will assume the relationships you show them to be causal (i.e. one causes the other). Example FT uses Inflation and unemployment, income and life expectancy	Use where an item's position in an ordered list is more important than its absolute or relative value. Don't be afraid to highlight the points of interest. Example FT uses Wash, digestion, league tables, constituency election results	Show values in a dataset and how often they occur. The shape for 'skew' of a distribution can be a memorable way of highlighting the lack of uniformity or equality in the data. Example FT uses Income distribution, population, GDP distribution, revealing inequality	Give emphasis to changing trends. Movements or extended series traversing decades or centuries. Choosing the correct time period is important to provide suitable context for the reader. Example FT uses Share price movements, economic time series, sectoral changes in a market	Show size comparisons. These can be relative (just being able to see the reader's interest is solely in the size of this component), consider a 'counted' number (for example, barrels, dollars or people) rather than a calculated rate or per cent. Example FT uses Commodity production, market capitalisation, volumes in general	Show how a single entity can be broken down into its component elements, if the reader's interest is solely in the size of this component, consider a magnitude-type chart instead. Example FT uses Fiscal budgets, company structures, national election results	Aside from locator maps only used when precise locations or geographical patterns in data are more important to the reader than anything else. Example FT uses Population density, natural resource locations, natural disaster risk/impact, catchment areas, variation in election results	Show the reader volumes or intensity of movement between two or more states or conditions. These might be logical sequences or geographical locations. Example FT uses Movement of funds, trade, migrants, lawsuits, information, relationship graphs.
Diverging bar  A simple standard bar chart that can handle both negative and positive magnitude values.	Scatterplot  The standard way to show the relationship between two continuous variables, each of which has its own axis.	Ordered bar  Standard bar charts display the ranks of values much more easily when sorted into order.	Histogram  The standard way to show a changing time series. If data are irregular, consider markers to represent data points.	Line  The standard way to show a changing time series. If data are irregular, consider markers to represent data points.	Column  The standard way to compare the size of things. Must always start at 0 on the axis.	Stacked column/bar  A simple way of showing part-to-whole relationships but can be difficult to read with more than a few components.	Basic choropleth (rate/ratio)  The standard approach for putting data on a map - should always be used rather than totals and use a sensible base geography.	Sankey  Shows changes in flows from one condition to at least one other: good for tracing the eventual outcome of a complex process.
Diverging stacked bar  Perfect for presenting survey results which involve sentiment (eg disagree/neutral/agree).	Column + line timeline  A good way of showing the relationship between an amount (column) and a rate (line).	Ordered column  See above.	Dot plot  A simple way of showing the change over time - but usually best with only one series of data at a time.	Column  Columns work well for showing change over time - but usually best with only one series of data at a time.	Bar  See above. Good when the data are not time series and labels have long category names.	Marimekko  A good way of showing the size and proportion of data at the same time - as long as the data are not too complicated.	Proportional symbol (count/magnitude)  Use for totals rather than rates - be wary that small differences in data will be hard to see.	Waterfall  Designed to show the sequencing of data through a flow process, typically budgets. Can include +/- components.
Spine  Splits a single value into two contrasting components (eg male/female).	Connected scatterplot  Usually used to show how the relationship between two variables has changed over time.	Ordered proportional symbol  Use when there are big variations between values and/or seeing fine differences between data is not so important.	Dot strip plot  Good for showing individual values in a distribution, can be a problem when too many dots have the same value.	Column + line timeline  A good way of showing the relationship over time between an amount (column) and a rate (line).	Paired column  As per standard column but allows for multiple series. Can become tricky to read when more than 2 series.	Pie  A common way of showing part-to-whole data - but be aware that it's difficult to accurately compare the size of the segments.	Flow map  For showing unambiguous movement across a map.	Chord  A complex but powerful diagram which can illustrate 2-way flows (and net winner) in a matrix.
Surplus/deficit filled line  The shaded area of these charts allows a balance to be shown - either against a baseline or between two series.	Bubble  Like a scatterplot, but adds additional detail by sizing the circles according to a third variable.	Dot strip plot  Dots placed in order on a strip are a space-efficient method of laying out ranks across multiple categories.	Barcode plot  Like dot strip plots, good for displaying all the data in a table, they work best when highlighting individual values.	Slope  Good for showing changing data as long as the data can be simplified into 2 or 3 categories.	Paired bar  See above.	Donut  Similar to a pie chart - but the centre can be a good way of making space to include more information about the data (eg total).	Contour map  For showing areas of equal value on a map. Can use deviation colour schemes for showing +/- values.	Network  Used for showing the strength and inter-connectedness of relationships of varying types.
XY heatmap  A good way of showing the patterns between 2 categories of data, less effective at showing fine differences in amounts.	Connected scatterplot  Usually used to show how the relationship between two variables has changed over time.	Slope  Perfect for showing how ranks have changed over time or very between categories.	Boxplot  Summarise multiple distributions by showing the median (centre) and range of the data.	Area chart  Use with care - these are good at showing changes to total, but seeing changes in components can be very difficult.	Marimekko  A good way of showing the size and proportion of data at the same time - as long as the data are not too complicated.	Proportional symbol  Use when there are big variations between values and/or seeing fine differences between data is not so important.	Equalized cartogram  Converting each unit on a map to a regular and equally sized shape - good for representing voting regions with equal value.	Heat map  Grid-based data values mapped with an intensity colour scale. As choropleth map - but not snapped to an administrative unit.
Lollipop  Lollipops draw more attention to the data value than standard barplots and can also show rank and value effectively.	Violin plot  Similar to a box plot but more effective with complex distributions and data that cannot be summarised with simple average.	Population pyramid  A standard way of showing the age and sex breakdown of a population distribution, effectively, back to back histograms.	Cumulative curve  A good way of showing how unequal a distribution is: y axis is always cumulative frequency, x axis is always a measure.	Calendar heatmap  A great way of showing temporal patterns (daily, weekly, monthly) - at the expense of showing precision in quantity.	Isotype (pictogram)  Excellent solution in some instances - use only with whole numbers (do not slice off an arm to represent a decimal).	Voronoi  A way of turning points into areas - any point within each area is closer to the central point than any other centroid.	Scaled cartogram (value)  Stretching and skewing a map so that each area is sized according to a particular value.	Dot density  Used to show the location of individual events/locations - make sure to annotate any patterns the reader should see.
Bump  Effective for showing changing rankings across multiple dates. For large datasets, consider grouping lines using colour.	Frequency polygons  For displaying multiple distributions of data. Like a regular line chart, best limited to a maximum of 3 or 4.	Boxplot  Use to emphasise individual points in a distribution. Points can be sized to an additional variable. Best with medium-sized datasets.	Cumulative curve  A good way of showing how unequal a distribution is: y axis is always cumulative frequency, x axis is always a measure.	Calendar heatmap  A great way of showing temporal patterns (daily, weekly, monthly) - at the expense of showing precision in quantity.	Isotype (pictogram)  Excellent solution in some instances - use only with whole numbers (do not slice off an arm to represent a decimal).	Gridplot  Good for showing % information, they work best when used on whole numbers and work well in small multiple layout forms.	Heat map  Grid-based data values mapped with an intensity colour scale. As choropleth map - but not snapped to an administrative unit.	Dot density  Used to show the location of individual events/locations - make sure to annotate any patterns the reader should see.
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Storytelling

Guiding users

- Selection of data + views
- Layout/composition
- Sequence + flow
- Highlighting
- Sequential reveal
- Animations
- Interactions
- Pre-selection of filters, queries, views
- Scrollytelling
- ...

Providing context

- Headline/caption
- Introductory text
- Annotations
- Related events
- Comparative measures
- Illustrations/pictures
- Callouts
- ...

Create understanding

- Legends
- Explanatory text
- Glossaries
- ...

Storytelling in data visualization should be a balance between hand-holding users and allowing for self-directed exploration as defined by how opinionated you want to be vs. just showing the data.

Dinner Break
until 7:20 pm

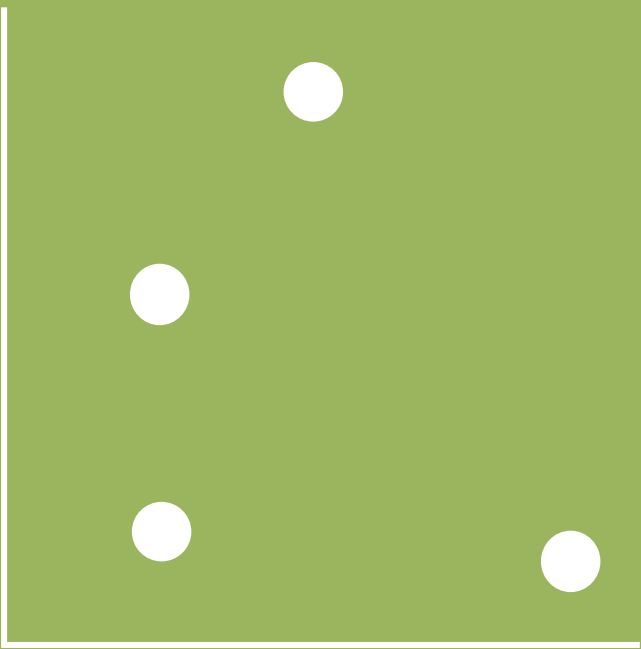
Context & Animation

Animation

Why?

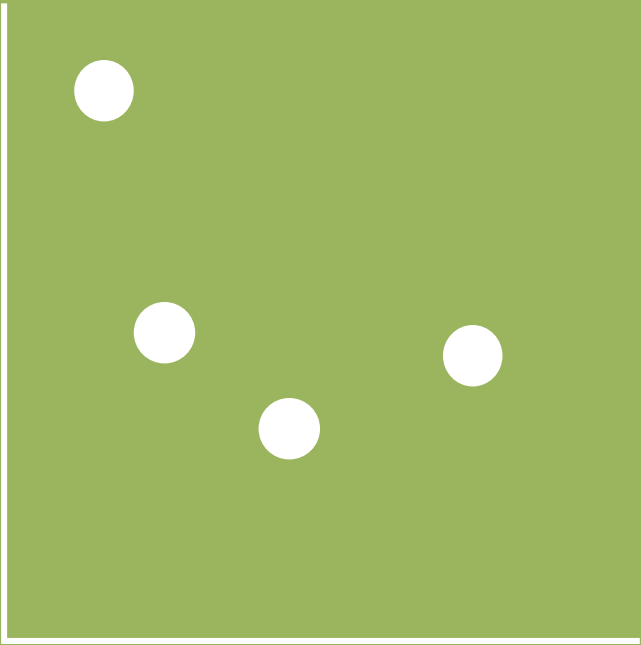
A

B



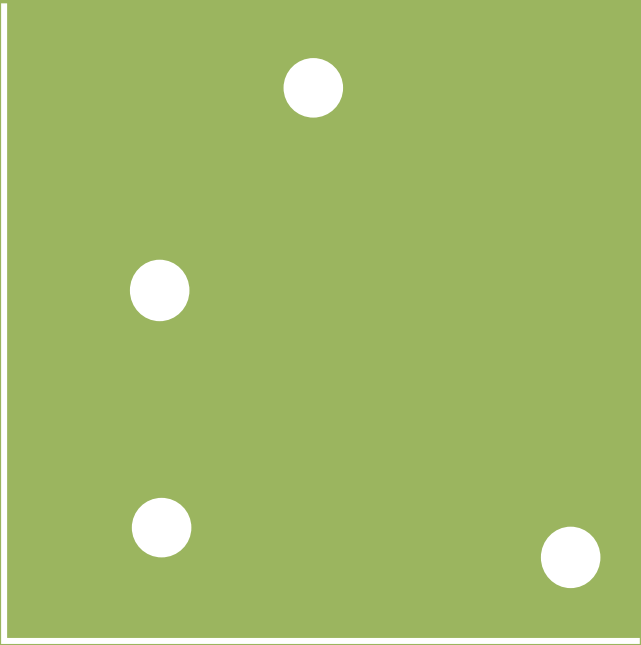
A

B



A

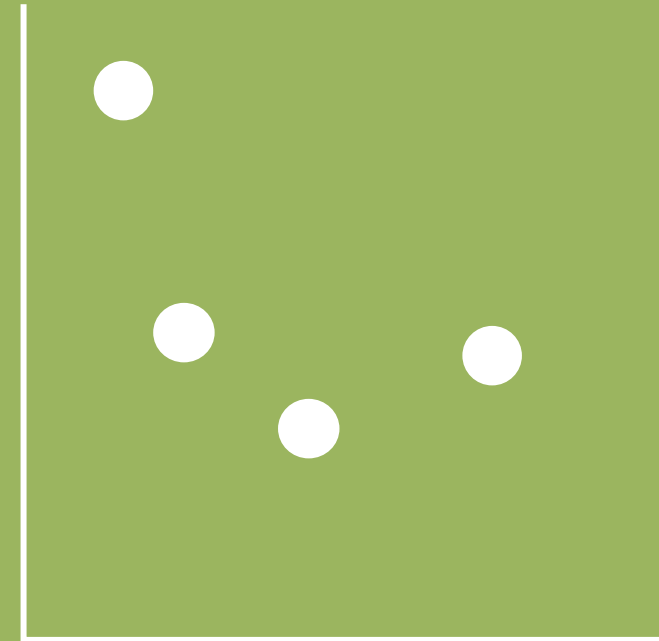
B



Help users maintain a sense of context between states.

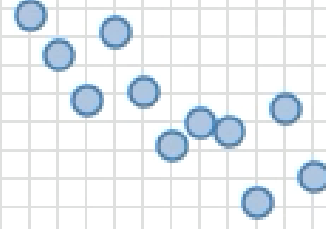
Better than tracking an item on their own using internal cognitive and memory resources.

Support better graphical perception of change when designed well.



What?

Pan + Zoom



Axis Change

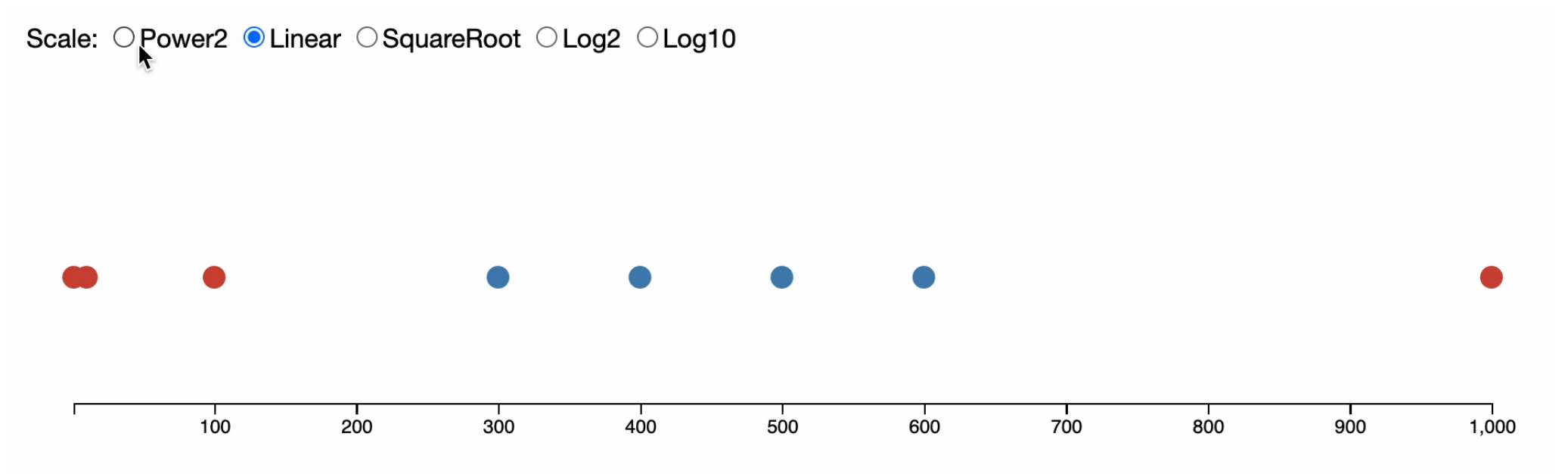


Chart Type

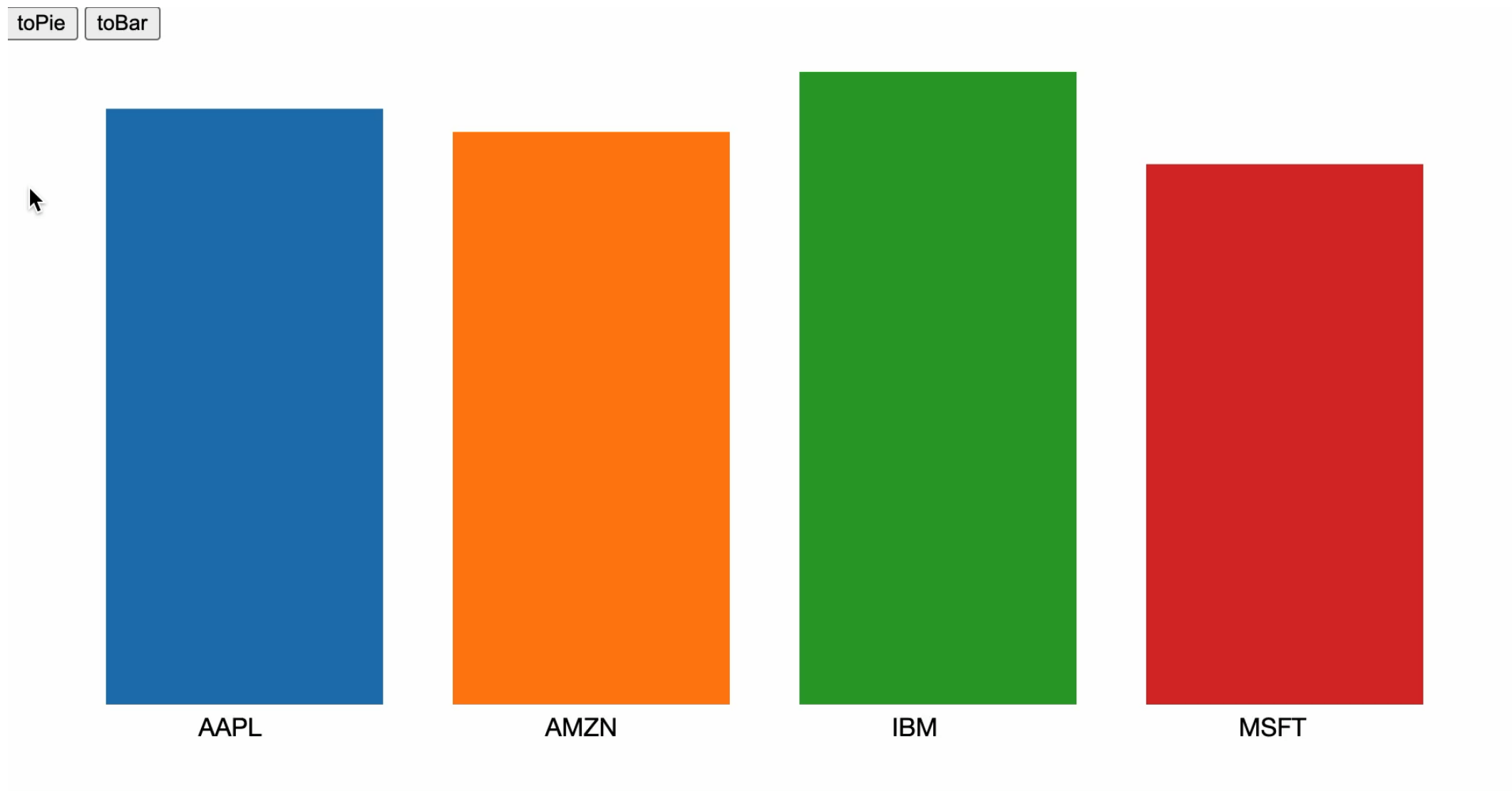
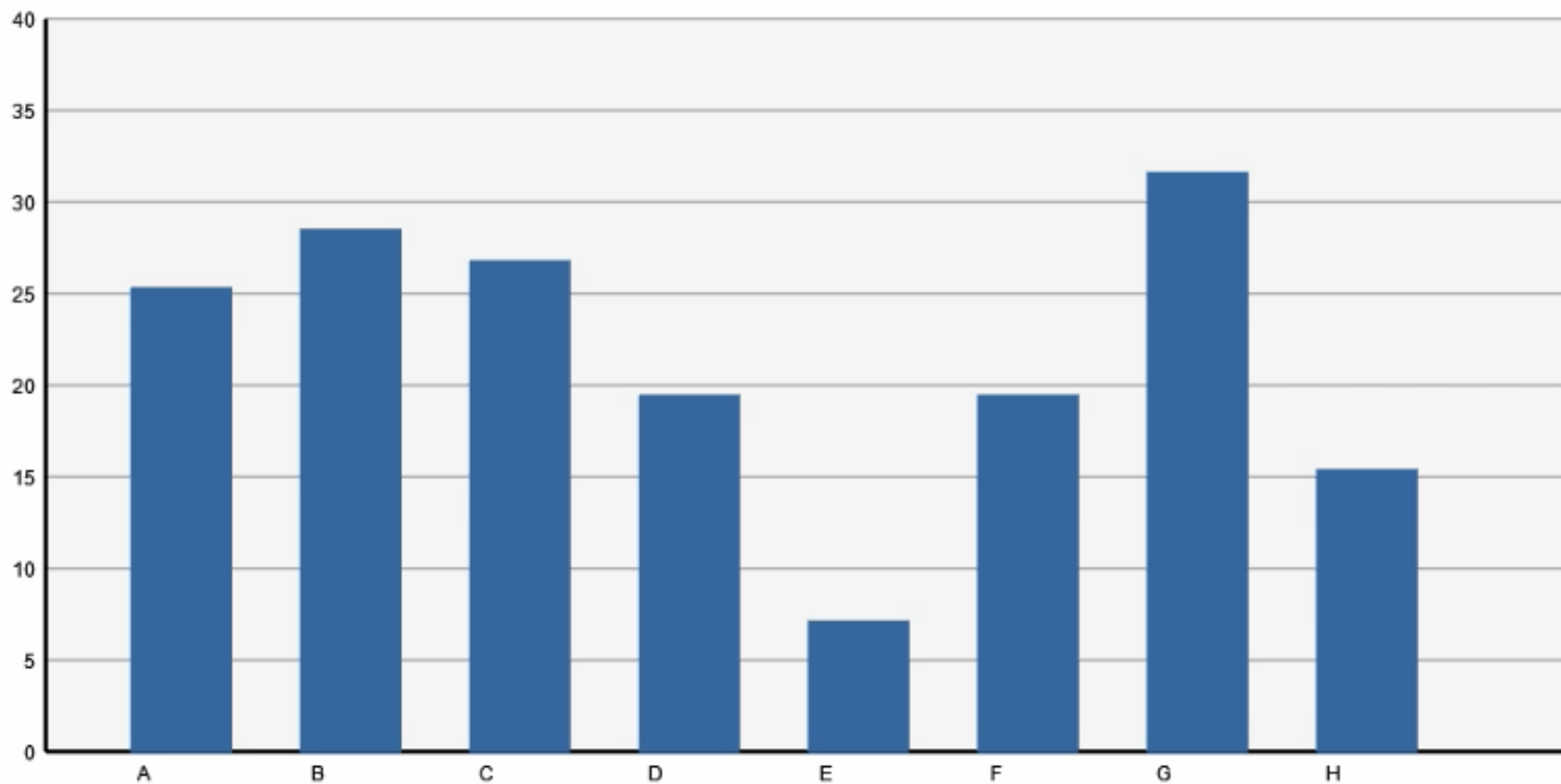


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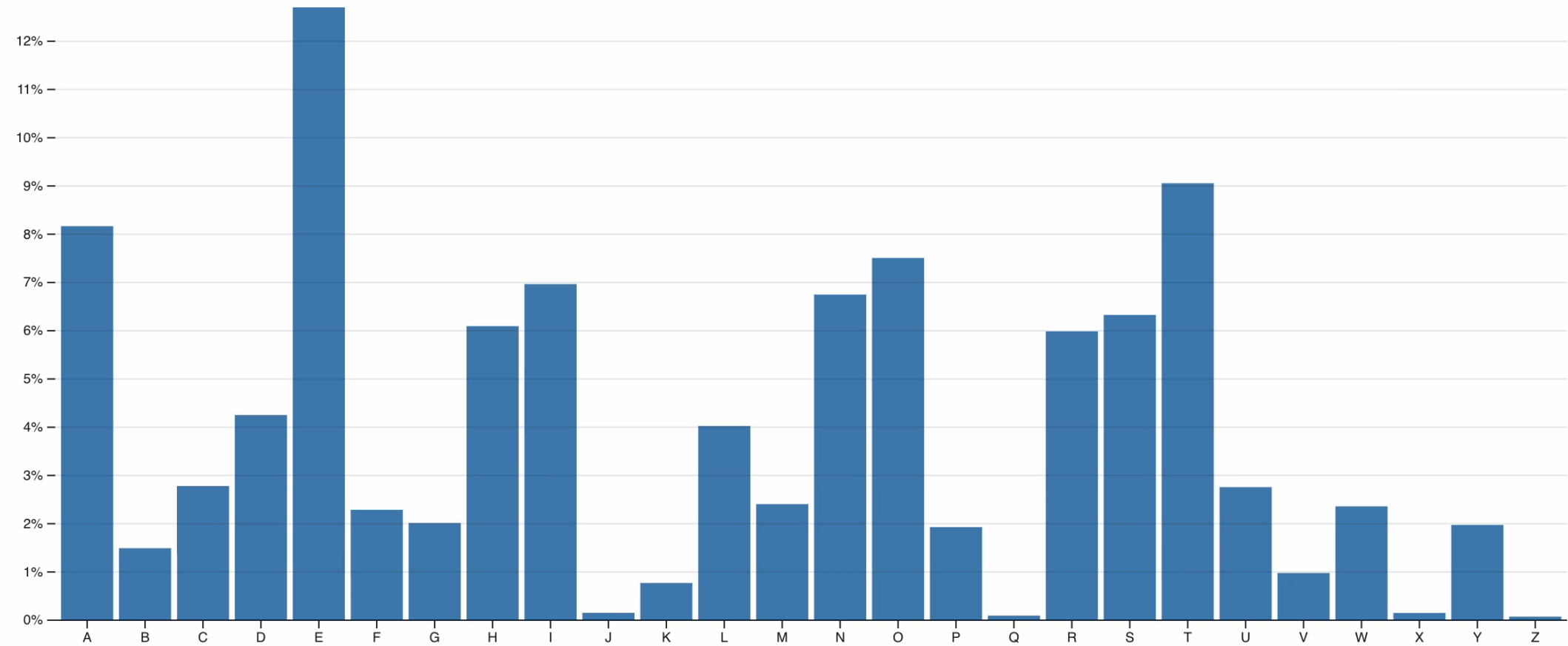


Order



Alphabetical

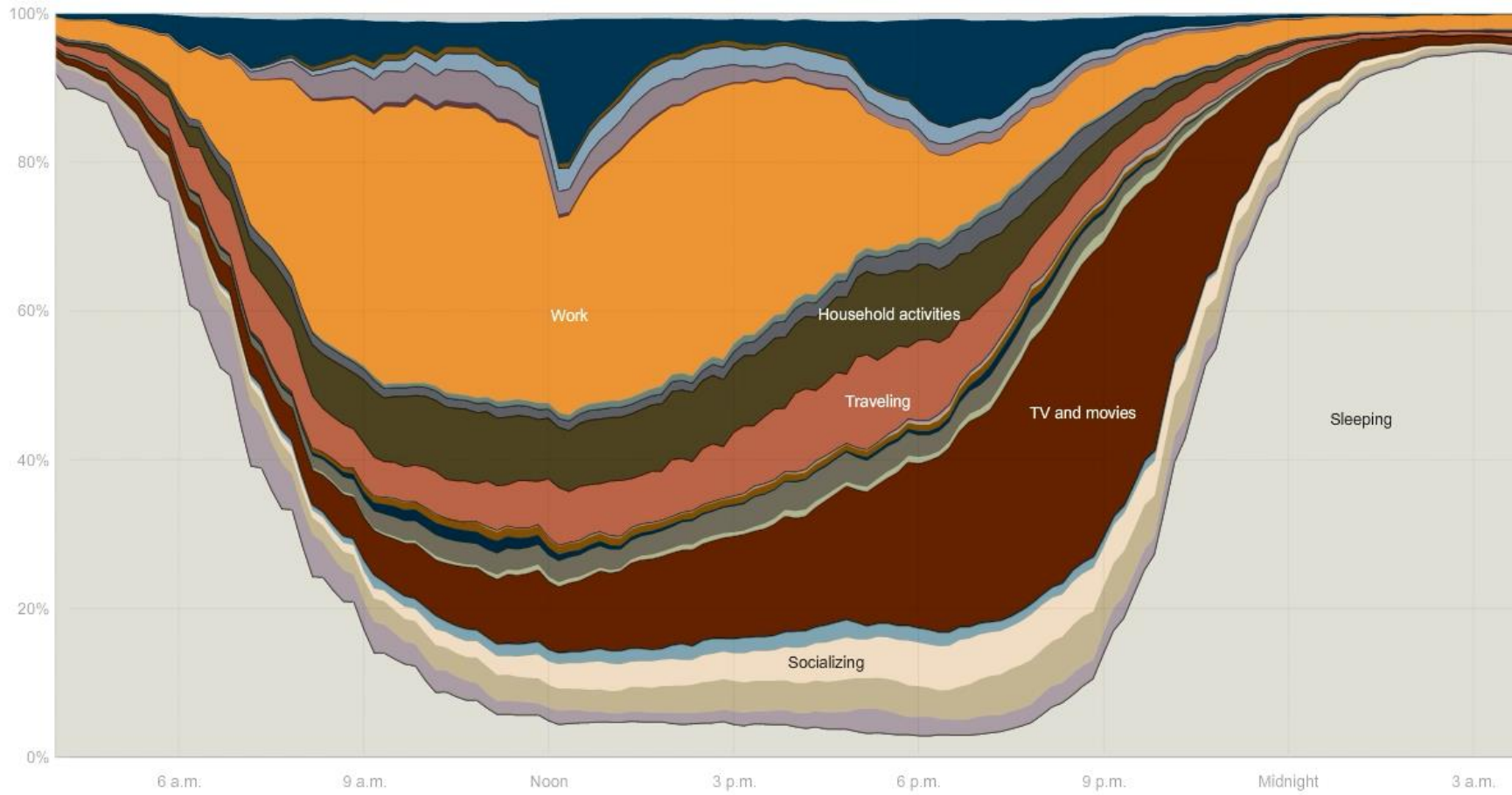
↑ Frequency



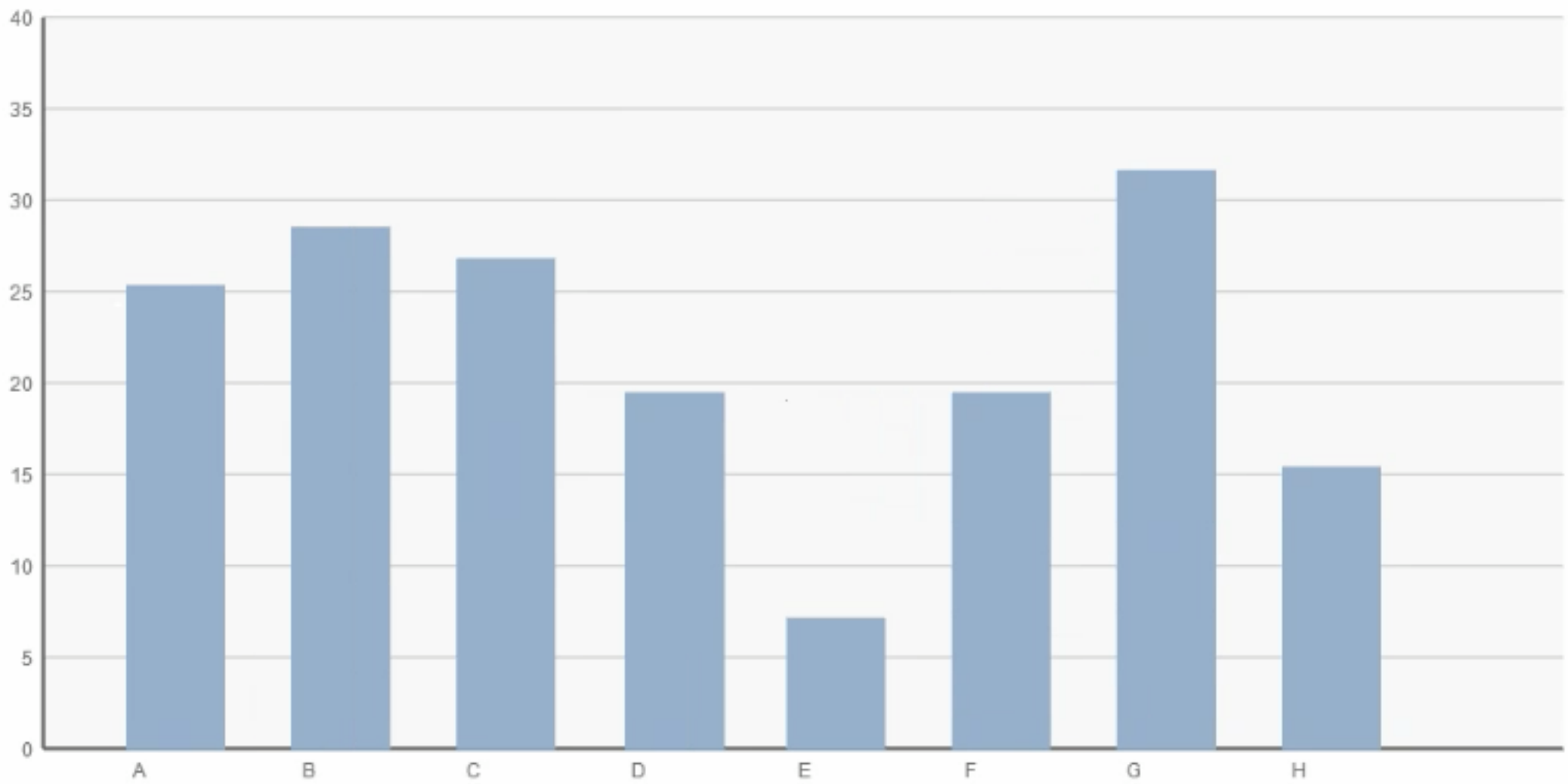
Reordering

Working peaks around 11 a.m., when 40 percent of men are at work. (On weekdays, 70 percent of men with a job are working at that time.)

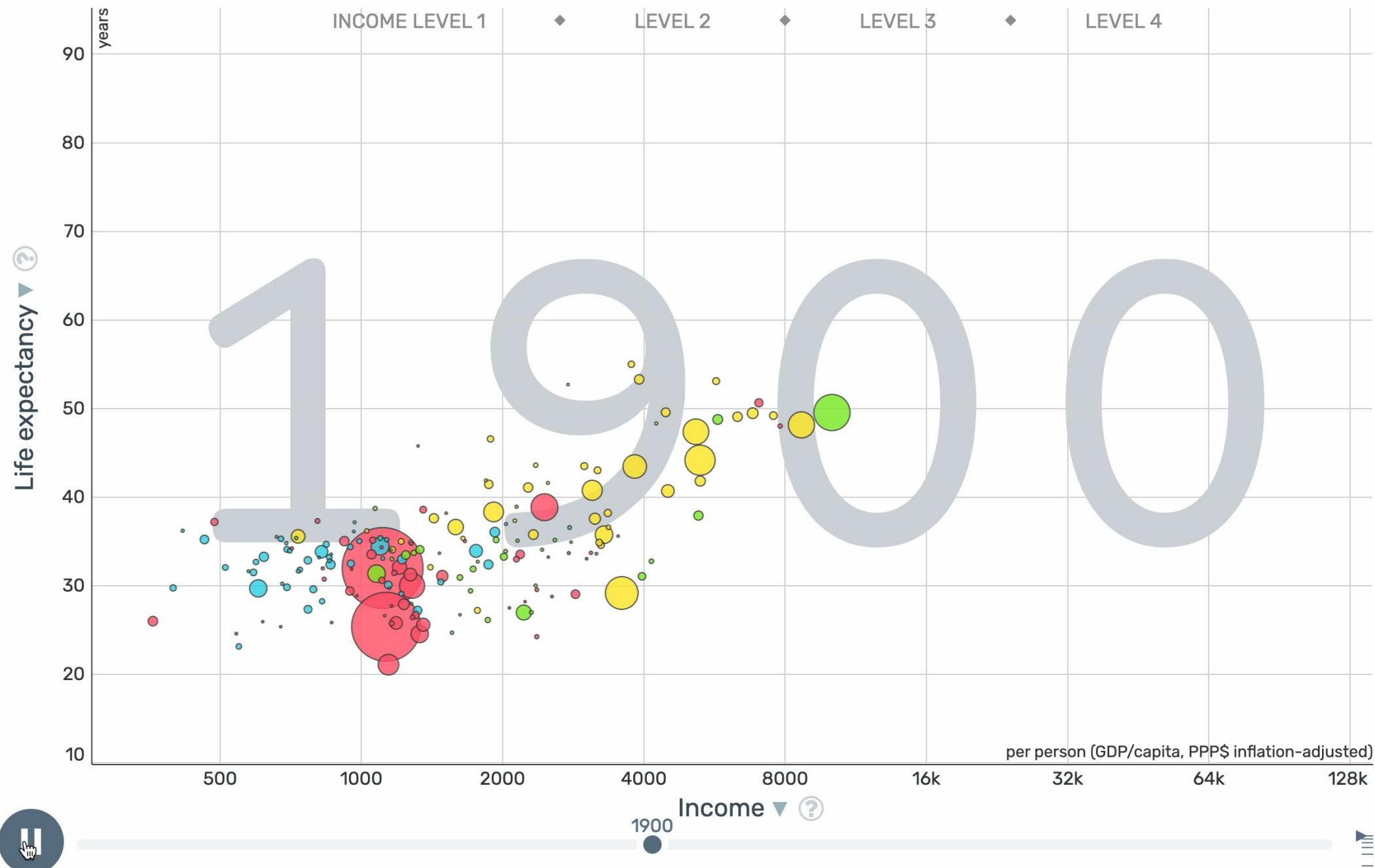
Everyone	Employed	White	Age 15-24	H.S. grads	No children
Men	Unemployed	Black	Age 25-64	Bachelor's	One child
Women	Not in lab...	Hispanic	Age 65+	Advanced	Two+ children



Adding/Removing



Changing Values



How?

The Illusion of Life

Disney Animation

Frank Thomas and Ollie Johnston

3. The Principles of Animation

"When we consider a new project, we really study it . . . not just the surface idea, but everything about it."
Walt Disney

A new jargon was heard around the studio. Words like "aiming" and "overlapping" and "pose to pose" suggested that certain animation procedures gradually had been isolated and named. Verbs turned into nouns overnight, as, for example, when the suggestion, "Why don't you stretch him out more?" became "Get more stretch on him." "Wow! Look at the squash on that drawing!" did not mean that a vegetable had splattered the artwork; it indicated that some animator had successfully shown a character in a flattened posture.

Some of this terminology was just assigning new meanings to familiar and convenient words. "Doing" a scene could mean acting out the intended movements, making exploratory drawings, or actually animating it; and once it was "done," the scene moved on to the next department. Layouts were done, backgrounds were done, recording was done, and, eventually, the whole picture had been done. Mixed in with these terms were the new names and phrases with more obscure meanings.

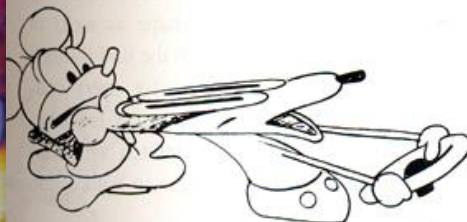
The animators continued to search for better methods of relating drawings to each other and had found a few ways that seemed to produce a predictable result. They could not expect success every time, but these special techniques of drawing a character in motion did offer some security. As each of these processes acquired a name, it was analyzed and perfected and talked about, and when new artists joined the staff

they were taught these practices as if they were the rules of the trade. To everyone's surprise, they became the fundamental principles of animation:

1. Squash and Stretch
2. Anticipation
3. Staging
4. Straight Ahead Action and Pose to Pose
5. Follow Through and Overlapping Action
6. Slow In and Slow Out
7. Arcs
8. Secondary Action
9. Timing
10. Exaggeration
11. Solid Drawing
12. Appeal

SQUASH and **STRETCH**

By far the most important discovery was what we call Squash and Stretch. When a fixed shape is moved about on the paper from one drawing to the next, there is a marked rigidity that is emphasized by the movement. In real life, this occurs only with the most rigid shapes, such as chairs and dishes and pans. Anything composed of living flesh, no matter how bony, will



Squash & Stretch



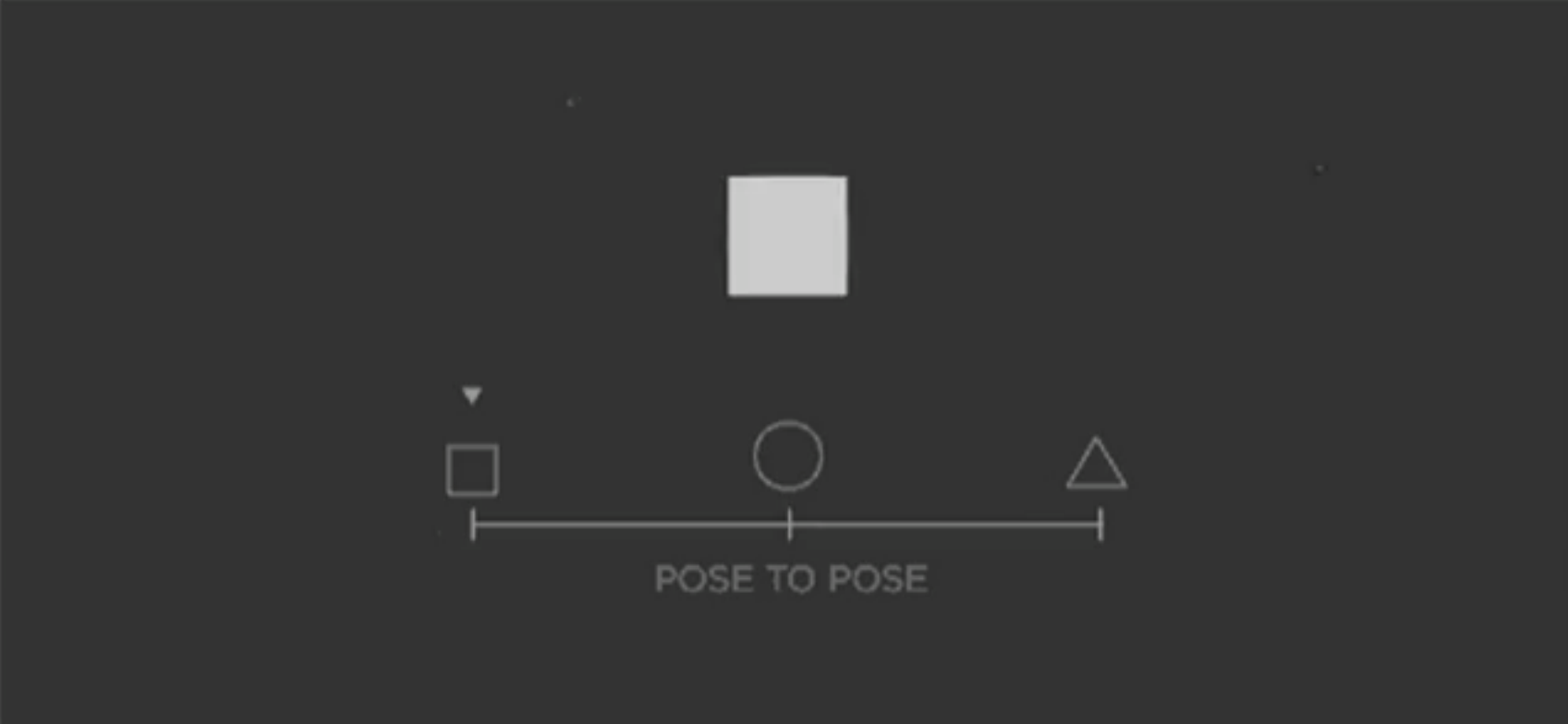
Anticipation



Staging



Pose to Pose



Follow Through



Slow In & Slow Out



Arcs



Secondary Action



Timing



Do one thing at a time.

Preserve valid mappings.

Preserve proportions.

Only necessary motion.

Make it meaningful.

Make it memorable.

Do one thing at a time.

Need to limit amount of change,
can't track everything at once
(max 4-5* objects).

Stagger/staging can help.

Or move groups of objects in
similar ways.

*Cavanagh and Alvarez (2005)

**Animation can be less accurate and/or slower.
But can also be more engaging and memorable.**

Robertson et al. (2008)

**Rosling's GapMinder talk is
predominantly an example
of successful storytelling,
which utilizes animation,
not the other way around.**

Presentation

(storytelling)

vs

Exploration

Context

**At best, isolated numbers tell you nothing.
At worst, they bias towards a very specific
conclusion.**

Total Revenue 2015

\$5.2

trillion

Includes local, state, and federal governments



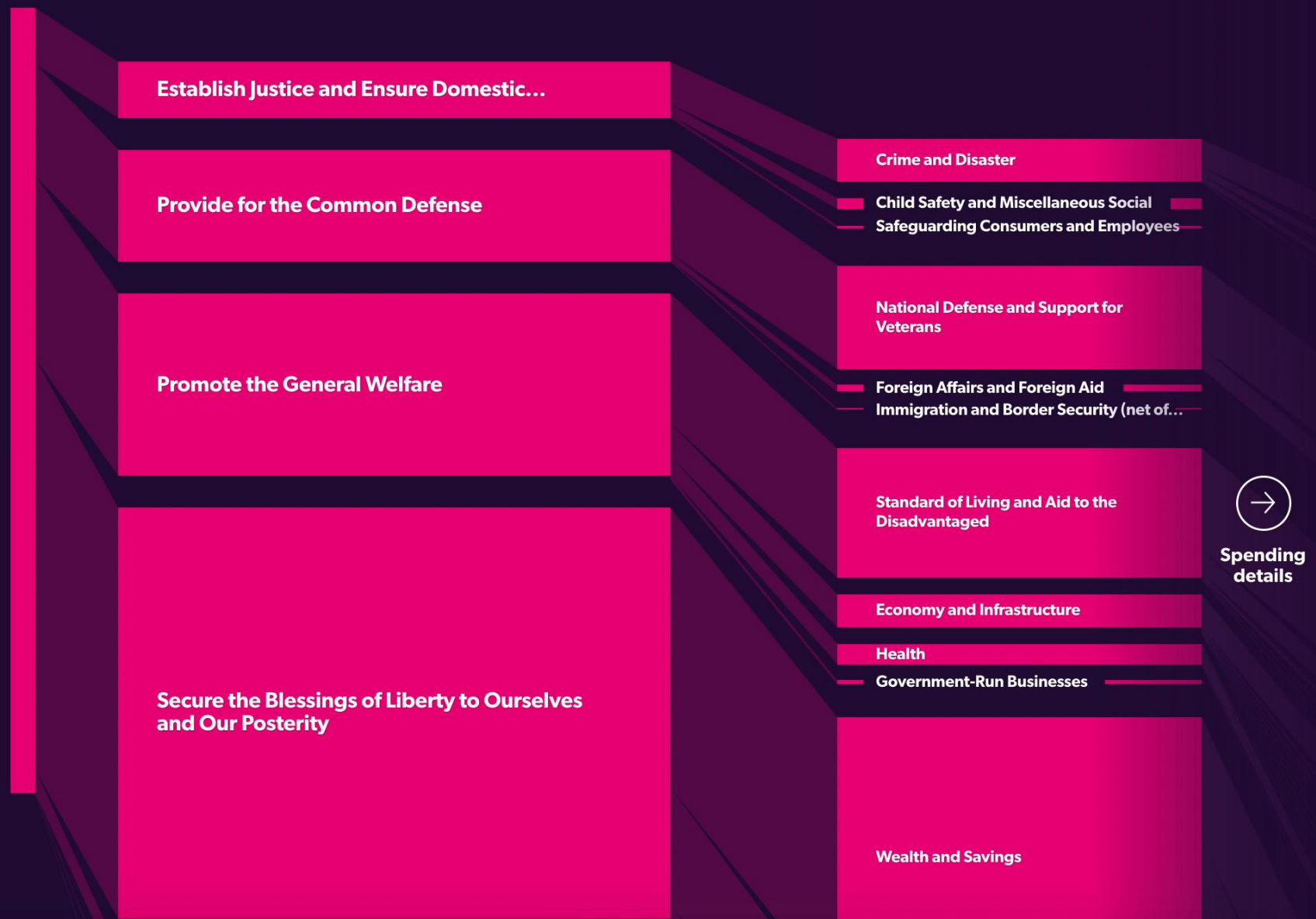
See where the money comes from

Total Spending 2015

\$5.7

trillion

Includes local, state, and federal governments



Spending details

Ball!!!

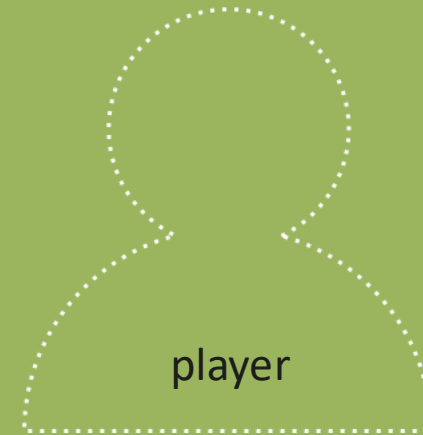


V=2.98 fps

hdg= 304°



Quantitative context





Historical context



Relational context

National Data ·

Economy

SEE ALL ISSUES

TAKE ACTION! →

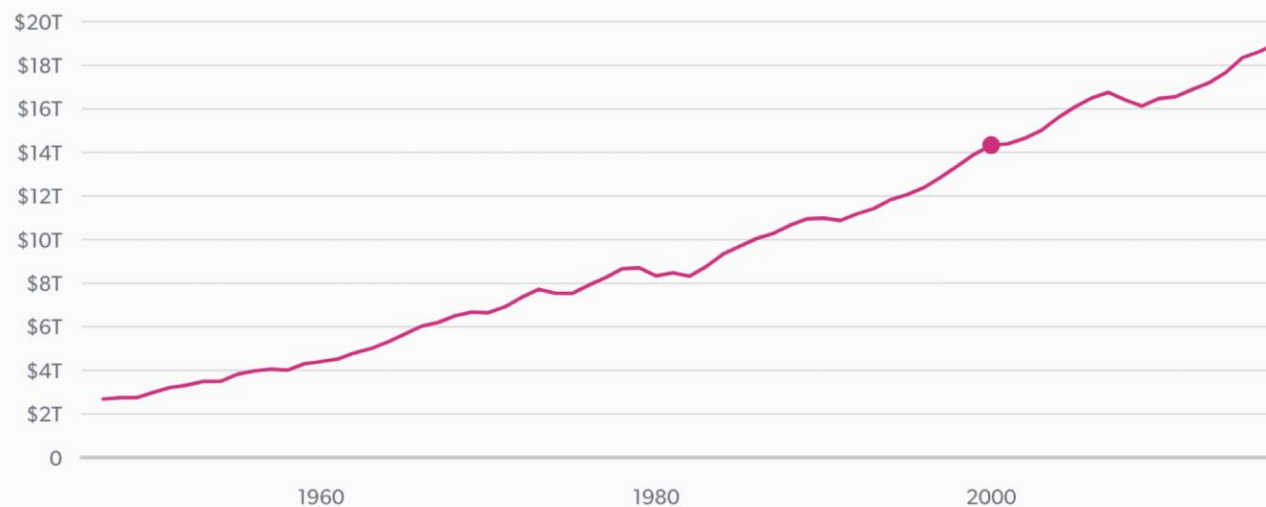
GDP

Table View

Gross domestic product *
\$14.33 Trillion

See more information >

2000



View source

Private Investment

Table View

The United States

2016

Total US population	323.41 million	↗
Total federal spending	\$3.88 trillion	↗
Gross domestic product	\$18.62 trillion	↗

FIND YOUR CANDIDATES

📍 Texas 11th District



SENATE

HOUSE



Beto O'Rourke

Democratic

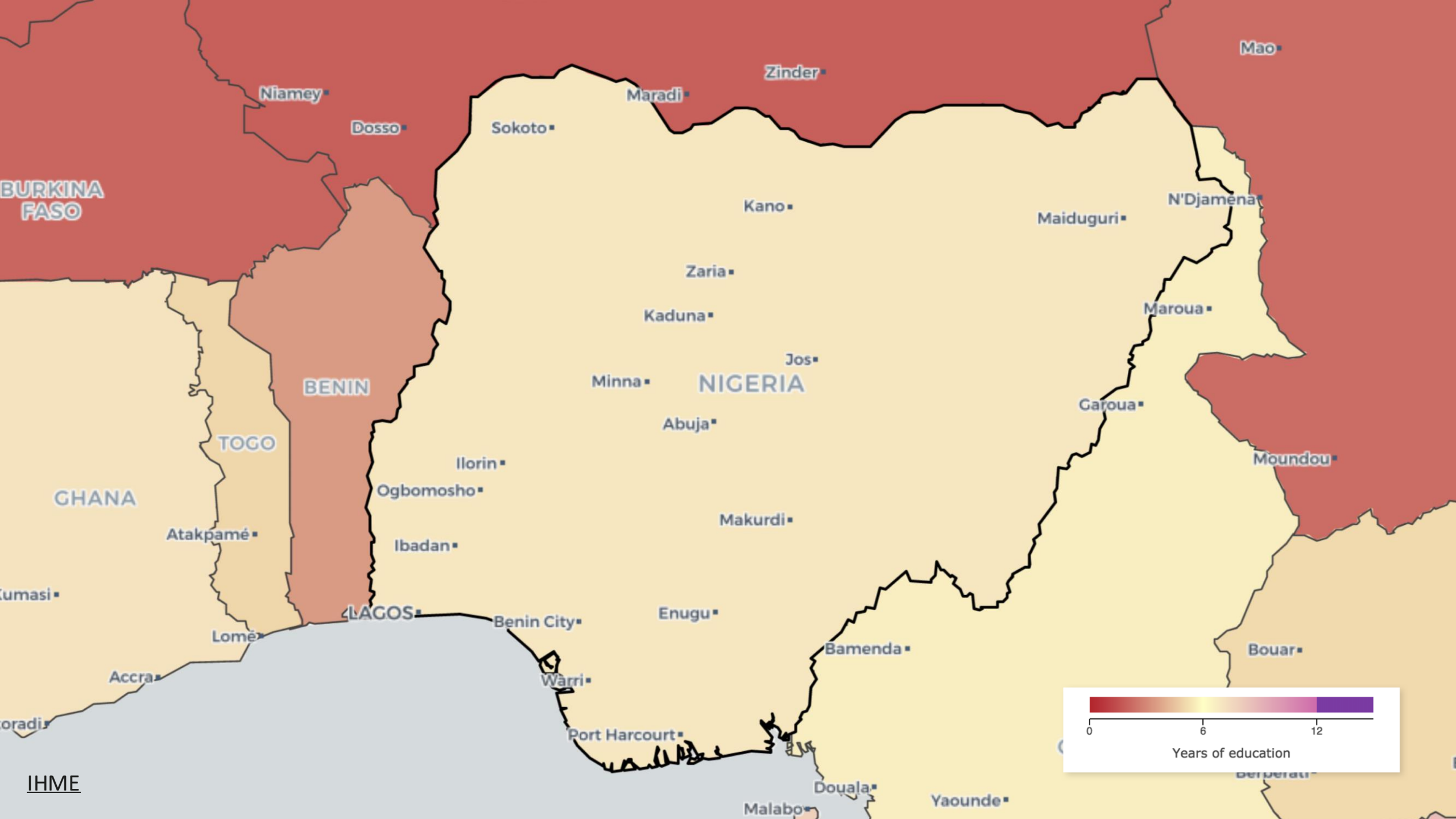


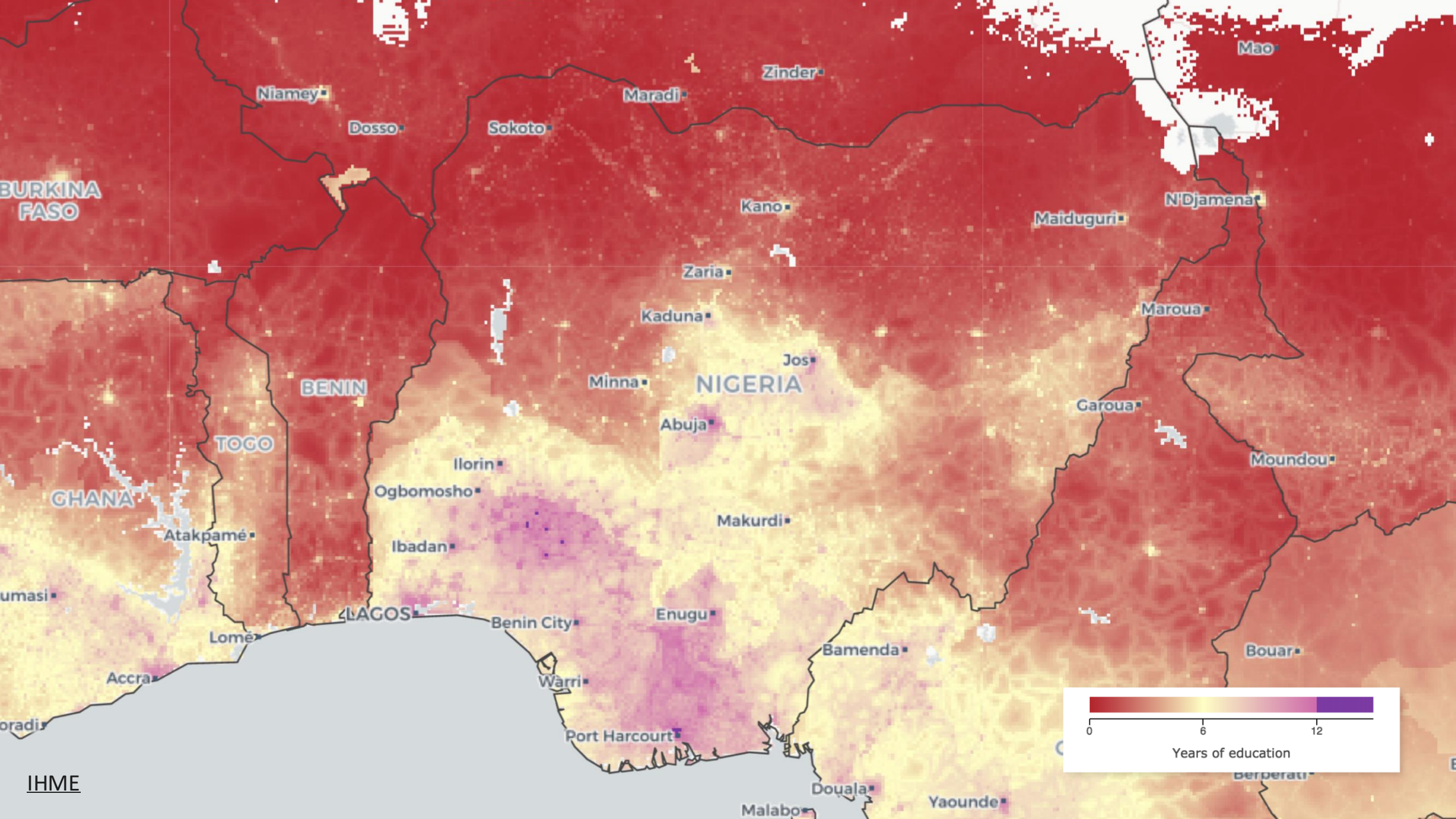
Everyone has a stake in prosperity. ☐

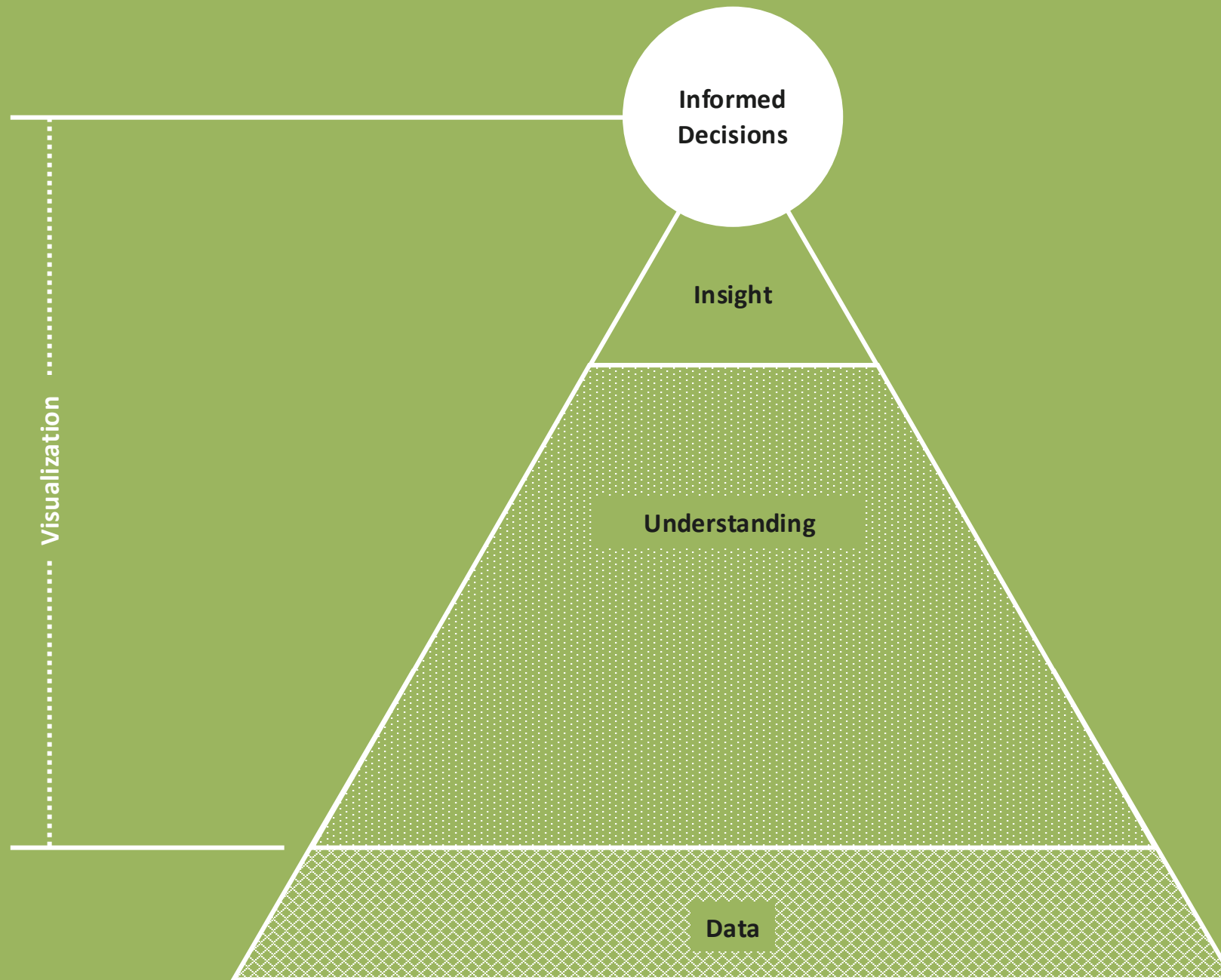
We have to promote fair and transparent markets that allow every Texan to compete and thrive. ☐

Lower the barrier to entry for small businesses so that every entrepreneur or emerging growth company can have an opportunity to be successful. ☐









Project

Feedback

Groups have asked for even more feedback!

Find your group's most recent report-out (slides, etc) or interactive visualization

Each group send a member to another to give feedback

Group

Time